



THE B.E.S.T. STANDARDS

Benchmarks for Excellent Student Thinking

Geometry EOC Assessment

2025 Released Test Items

1. Two statements about triangle JKL are shown.

- $m \angle L = 90^\circ$

- $\sin(J) = \frac{99}{101}$

Which ratio represents $\cos(K)$?

Ⓐ $\frac{101}{99}$

Ⓑ $\frac{99}{101}$

Ⓒ $\frac{20}{101}$

Ⓓ $\frac{20}{99}$

Geometry EOC

2. Line segment MN has endpoints $M (-7, 10)$ and $N (-1, 1)$. Point P is on line segment MN such that the ratio $MP:PN$ is $2:1$.

What is the x -coordinate of point P ?

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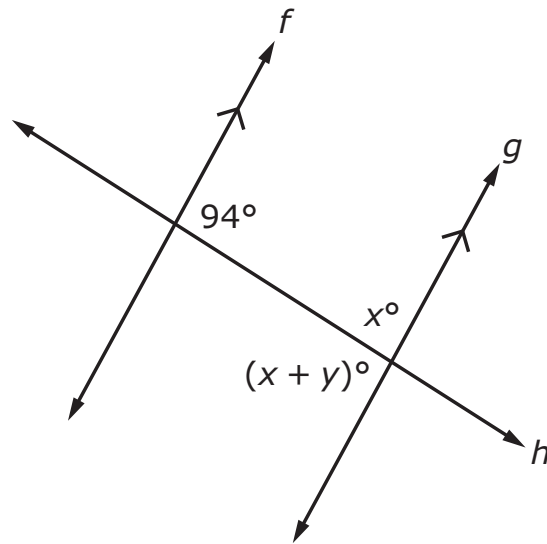
✖

1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\Box}{\Box}$

-
- 3.** Which statement is true when a three-dimensional figure is dilated by a given scale factor?
- Ⓐ A dilation by a scale factor of $\frac{2}{3}$ results in an increase in volume.
 - Ⓑ A dilation by a scale factor of 0.1 results in no change in volume.
 - Ⓒ A dilation by a scale factor of $\frac{3}{2}$ results in a decrease in surface area.
 - Ⓓ A dilation by a scale factor of 3.5 results in an increase in surface area.

Geometry EOC

4. Two parallel lines, f and g , are cut by a transversal, h , as shown.



What are the values of x and y ?

$x =$

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1	2	3		
4	5	6		
7	8	9		
	0			
.	-	$\frac{\Box}{\Box}$		

$y =$

				
1	2	3		
4	5	6		
7	8	9		
	0			
.	-	$\frac{\square}{\square}$		

5. Which statement must be true for all parallelograms?

- Ⓐ Diagonals are congruent.
- Ⓑ Diagonals are perpendicular.
- Ⓒ Opposite angles are complementary.
- Ⓓ Consecutive angles are supplementary.

6. A true statement is given.

"If two given shapes are circles, then the shapes are similar."

Select the boxes to identify what could be the converse, inverse, and contrapositive of the given statement.

	Converse	Inverse	Contrapositive
If two given shapes are not similar, then the shapes are not circles.	(A)	(B)	(C)
If two given shapes are similar, then the shapes are circles.	(D)	(E)	(F)
If two given shapes are not circles, then the shapes are not similar.	(G)	(H)	(I)

7. This question has **two** parts.

The vertices of two triangles are given.

- The coordinates of the vertices of the first triangle are $Q (2, 0)$, $R (6, 0)$, and $S (2, 4)$.
- The coordinates of the vertices of the second triangle are $T (8, 0)$, $U (4, -4)$, and $V (4, 0)$.

Part A

Which statement describes a relationship between the first and second triangles?

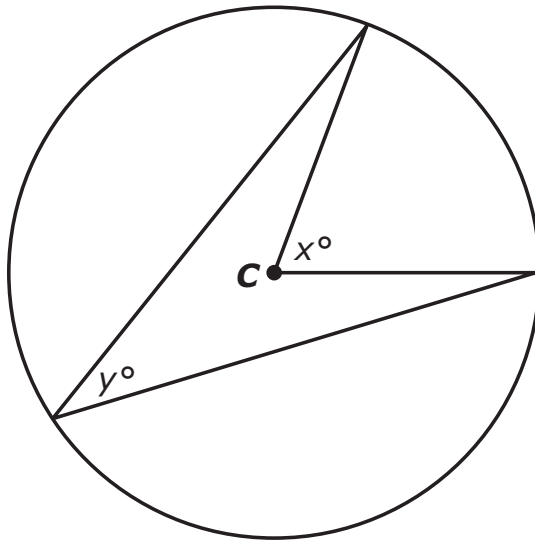
- Ⓐ $\triangle QRS \cong \triangle TUV$
- Ⓑ $\triangle QRS \cong \triangle VTU$
- Ⓒ $\triangle QRS \sim \triangle TUV$
- Ⓓ $\triangle QRS \sim \triangle UVT$

Part B

Select expressions to create a transformation that proves the statement in Part A.

$$(x, y) \rightarrow ([\text{Ⓐ } 2x \text{ Ⓑ } -x \text{ Ⓒ } x + 2 \text{ Ⓓ } x - 2], [\text{Ⓐ } 2y \text{ Ⓑ } -y \text{ Ⓒ } y + 8 \text{ Ⓓ } y - 8])$$

- 8.** An inscribed angle and a central angle are shown in circle C .



Create an equation to show the relationship between x and y .

$\leftarrow$	$\rightarrow$	$\curvearrowright$	$\curvearrowleft$	$\times$										
1	2	3	x	y										
4	5	6	+	-	$\bullet$	$\div$								
7	8	9	<	$\leq$	=	$\geq$	>							
	0		$\square^\square$	$\square_\square$	()		$\sqrt{\square}$	$\sqrt[\square]{\square}$	π	i				
.	-	$\frac{\square}{\square}$	sin	cos	tan	arcsin	arccos	arctan						

Geometry EOC

9. Figure *A* is drawn on a coordinate plane. Transformations are performed on figure *A* to create figure *B*.

Select the boxes to identify whether each transformation performed on figure *A* results in a congruent or noncongruent figure *B*.

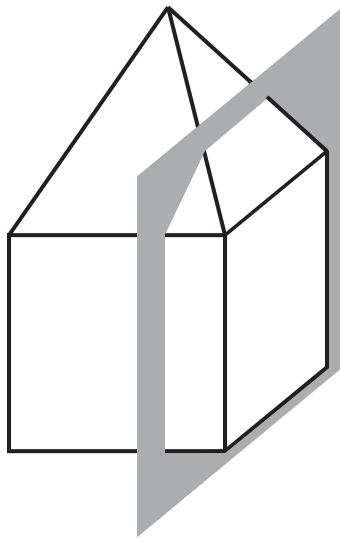
	Congruent	Noncongruent
$(x, y) \rightarrow (y, -x)$	☐ Ⓐ	☐ Ⓑ
$(x, y) \rightarrow \left(\frac{1}{2}x, \frac{1}{2}y\right)$	☐ Ⓒ	☐ Ⓓ
a reflection across the line $y = -x$	☐ Ⓔ	☐ Ⓕ
a 50-degree clockwise rotation about the origin	☐ Ⓖ	☐ Ⓖ

10. A line segment has an endpoint at $(1, 1)$ and a midpoint at $(1, -3)$.

What are the coordinates of the other endpoint?

- Ⓐ $(1, 5)$
- Ⓑ $(1, -1)$
- Ⓒ $(1, -5)$
- Ⓓ $(1, -7)$

- 11.** The figure consisting of a right pyramid and a right rectangular prism is sliced perpendicular to its base, as shown.



What is the shape of the resulting cross section?

- Ⓐ a rectangle
- Ⓑ a trapezoid
- Ⓒ a composite shape made up of a rectangle and a triangle
- Ⓓ a composite shape made up of a rectangle and a trapezoid

12. A triangle has vertices at $T(2, 5)$, $U(6, 6)$, and $V(4, -3)$.

Complete the statements to explain why triangle TUV is a right triangle.

The slope of $\overline{TU}$ is

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7	8	9
	0	
.	-	$\frac{\square}{\square}$

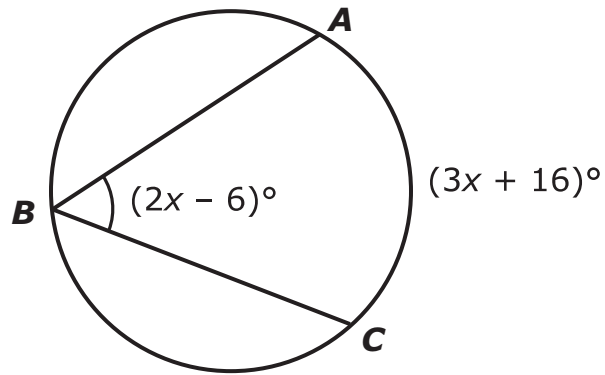
and the slope of $\overline{TV}$ is

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1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\square}{\square}$

Triangle TUV is a right triangle because $\overline{TU}$ and $\overline{TV}$ are
 [Ⓐ skewed Ⓑ parallel Ⓒ perpendicular] and therefore form a right angle.

13. Points A , B , and C lie on the circle, as shown.



- The measure of $\angle ABC$ is $(2x - 6)^\circ$.
- The measure of $\widehat{AC}$ is $(3x + 16)^\circ$.

What is the measure, in degrees, of $\angle ABC$?

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1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\Box}{\Box}$

- 14.** A cone has a radius of 3 units, a height of 4 units, and a slant height of 5 units.

Which expression represents the surface area, in square units, of the cone?

- Ⓐ 15π
- Ⓑ 21π
- Ⓒ 24π
- Ⓓ 29π

Geometry EOC

15. Rectangle $ABCD$ has the given side lengths.

- $AB = 5$ units
- $BC = 11$ units

Rectangle $PQRS$ is congruent to rectangle $ABCD$.

What is the length of $\overline{QR}$, in units?

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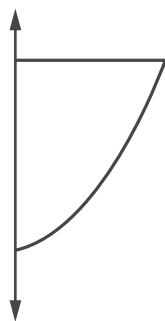
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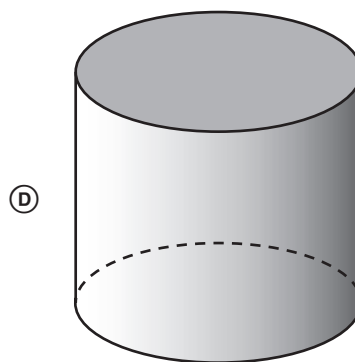
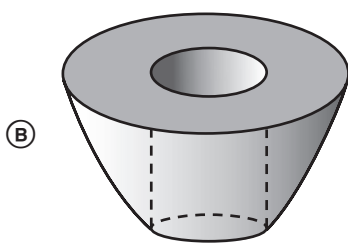
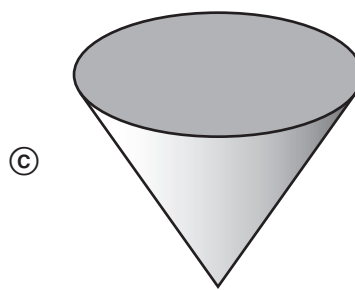
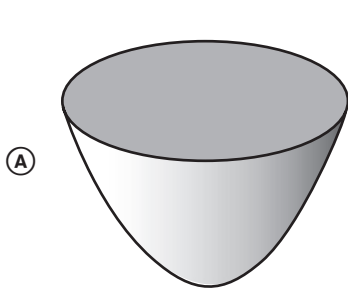
1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\Box}{\Box}$

16. A two-dimensional figure and a vertical line are shown.

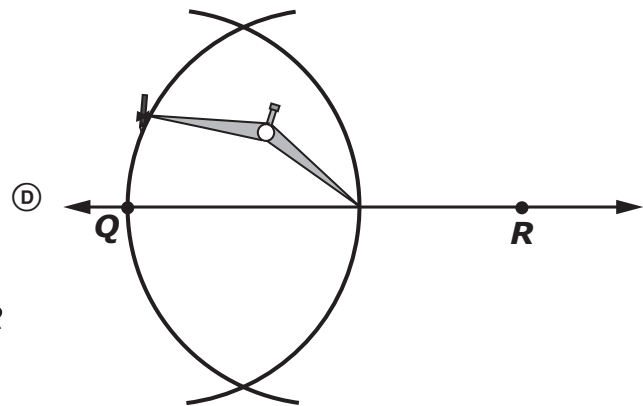
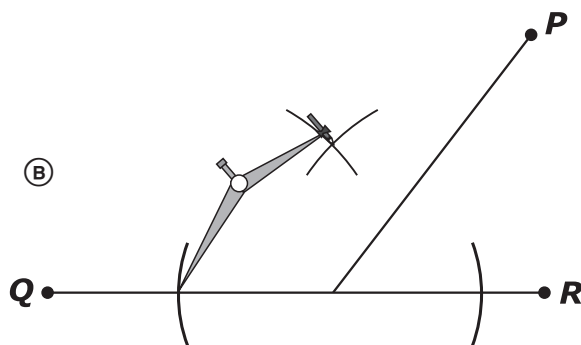
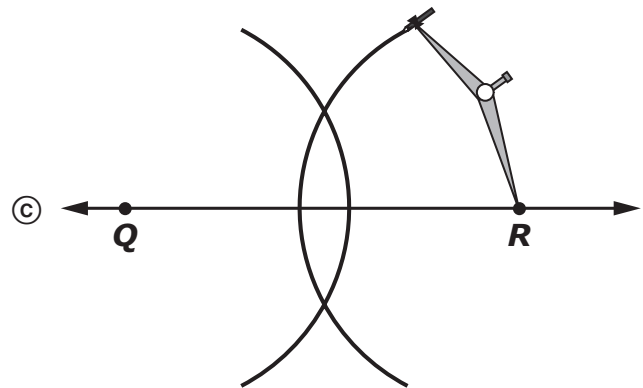
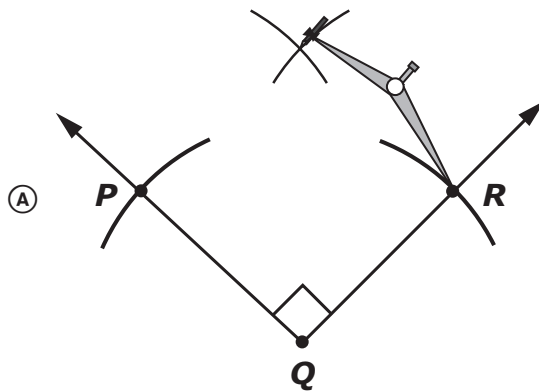


The figure is rotated about the vertical line.

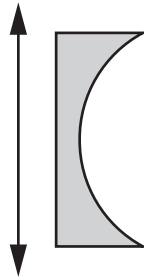
Which object could be the result of this rotation?



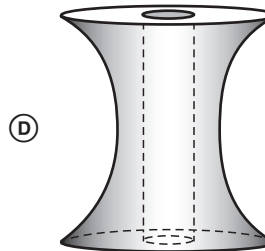
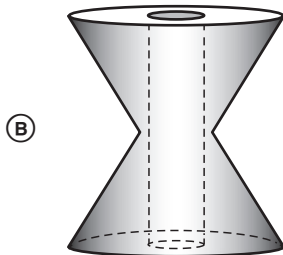
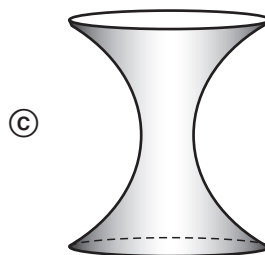
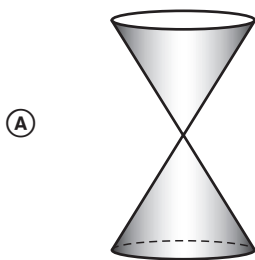
17. Which figure shows steps leading to the construction of a perpendicular bisector?



18. A figure and a vertical line are shown.



Which object is generated by rotating the figure around the vertical line?



Geometry EOC

- 19.** Triangle M has a base length of 18 inches and a height of 12 inches. Triangle N is congruent to triangle M .

What is the area, in square inches, of triangle N ?

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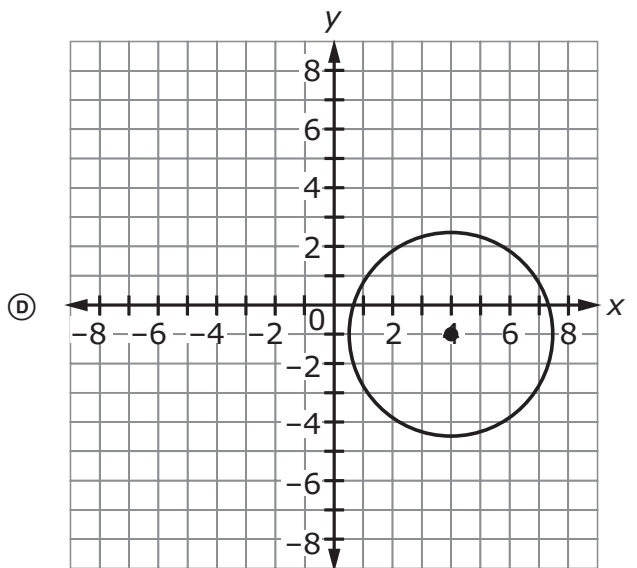
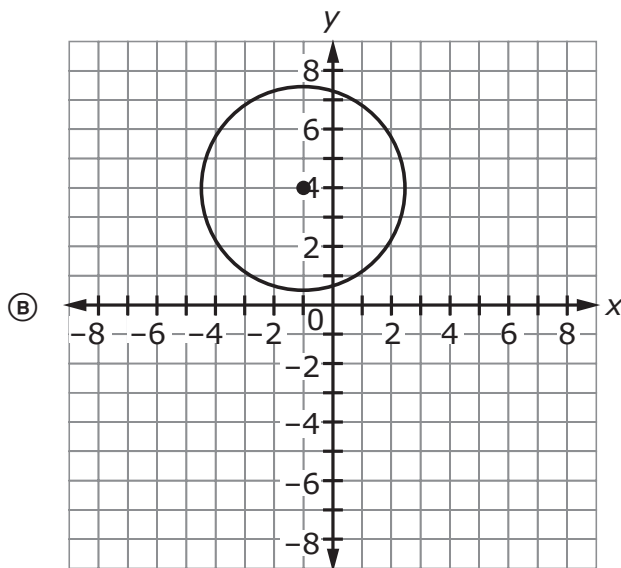
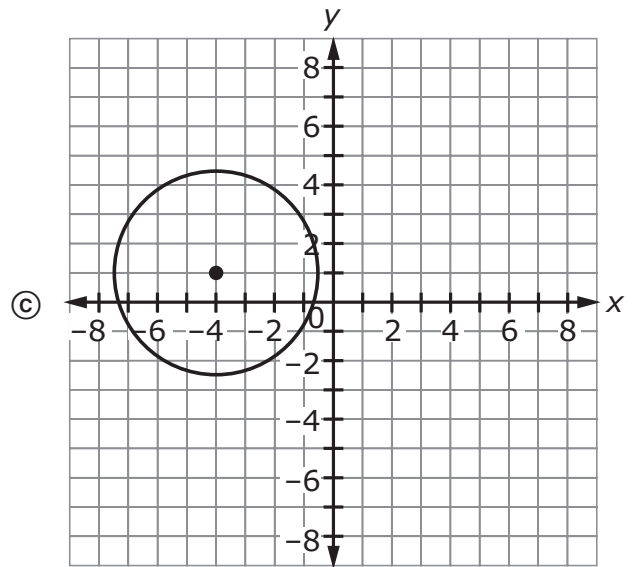
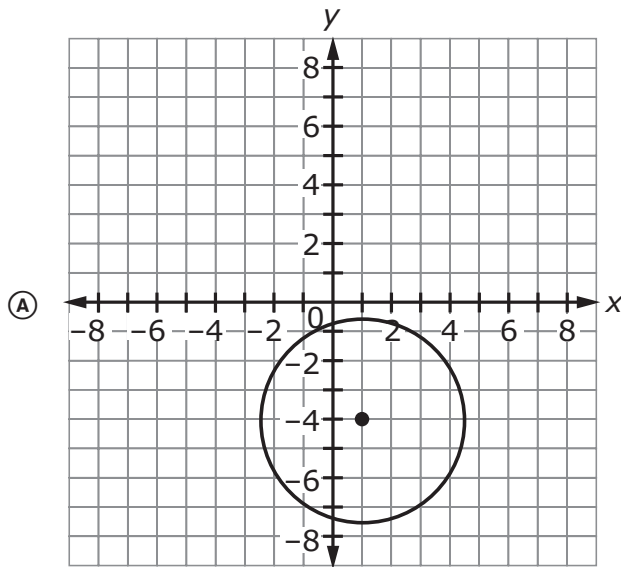
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1	2	3
4	5	6
7	8	9
	0	
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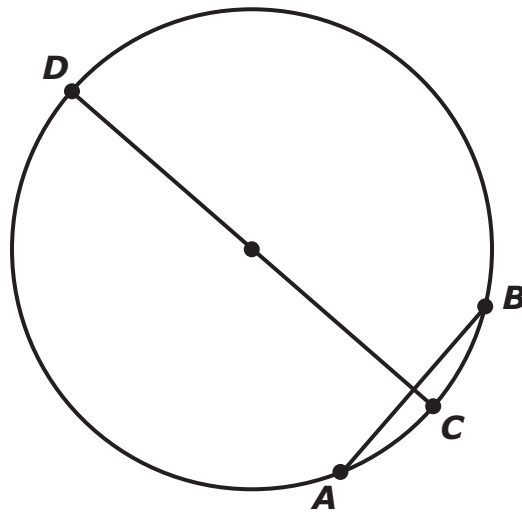
20. The equation of a circle is given.

$$(x - 1)^2 + (y + 4)^2 = 12.25$$

Which graph represents the equation?



21. A circle with chord AB and its perpendicular bisector CD is shown.



The measure of $\widehat{AC}$ is 28° .

What is the measure, in degrees, of $\widehat{BD}$?

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1	2	3
4	5	6
7	8	9
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22. Triangle XYZ is transformed to create triangle $X'Y'Z'$ using the rule $(x, y) \rightarrow (x + b, y + c)$.

- Triangle XYZ has the vertices $(3, -2)$, $(3, -4)$, and $(0, -4)$, respectively.
- Triangle $X'Y'Z'$ has the vertices $(-1, 5)$, $(-1, 3)$, and $(-4, 3)$, respectively.

What are the values of b and c ?

$b =$

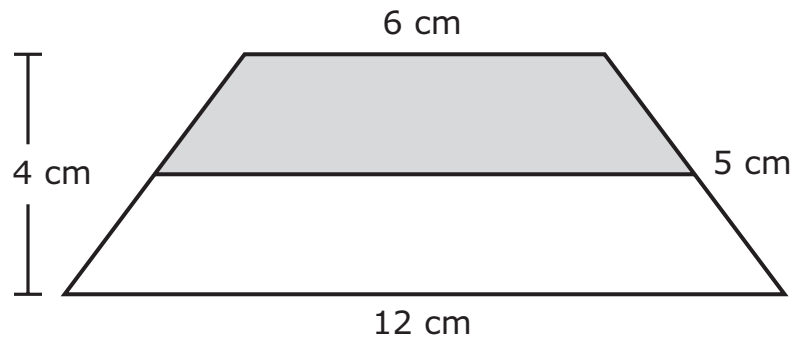
←	→	↶	↷
✖			
1	2	3	
4	5	6	
7	8	9	
	0		
.	-	$\frac{\Box}{\Box}$	

$c =$

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1	2	3	
4	5	6	
7	8	9	
	0		
.	-	$\frac{\Box}{\Box}$	

Geometry EOC

- 23.** A tile is in the shape of an isosceles trapezoid. The tile is divided into 2 parts by its midsegment. The trapezoid, with lengths in centimeters (cm), is shown.

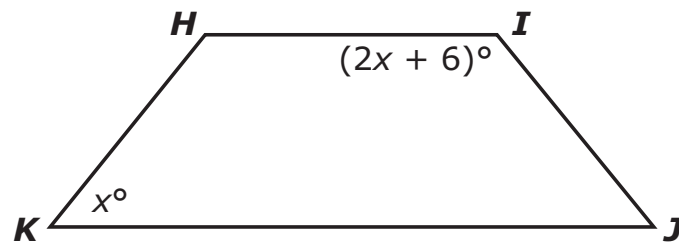


The top part is shaded.

What is the area, in square centimeters, of the shaded part?

- Ⓐ 12
- Ⓑ 15
- Ⓒ 18
- Ⓓ 36

24. An isosceles trapezoid is shown.



What is the measure of $\angle HIJ$, in degrees?

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1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\square}{\square}$

Geometry EOC

- 25.** Rectangle G has a length of 15 inches and a width of 20 inches.
Rectangle H is similar to rectangle G and has a length of 18 feet.

What is the width, in feet, of rectangle H ?

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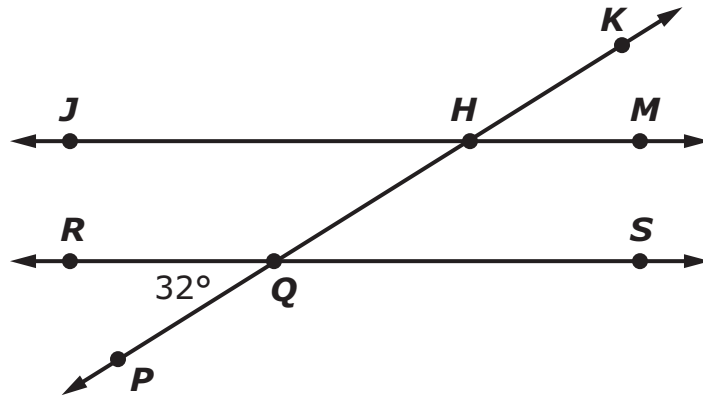
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1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\Box}{\Box}$

26. Transversal line KP passes through parallel lines JM and RS , as shown.



Complete the sentences to describe angle QHJ .

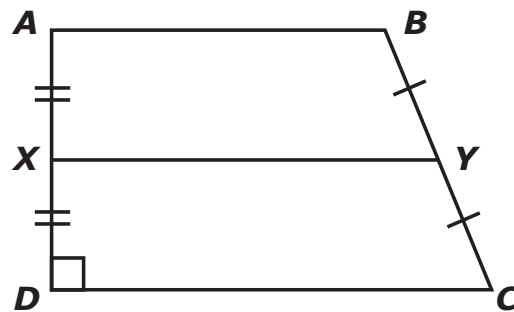
Angles PQR and QHJ are [☐ A congruent ☐ B supplementary ☐ C complementary] because they are [☐ A consecutive ☐ B corresponding ☐ C alternate exterior] angles.

The measure of angle QHJ is

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1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\square}{\square}$

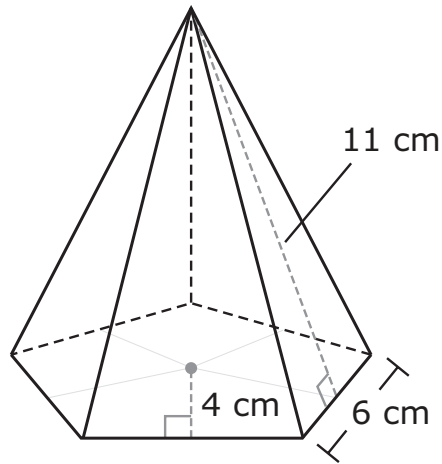
27. Trapezoid $ABCD$ is shown.



Select all the true statements.

- Ⓐ $\overline{AX} \cong \overline{BY}$
- Ⓑ $\overline{DX} \cong \overline{CY}$
- Ⓒ $\overline{XY} \parallel \overline{CD}$
- Ⓓ $XY = \frac{1}{2}(AB + CD)$
- Ⓔ trapezoid $ABYX \cong$ trapezoid $XYCD$

- 28.** A regular pentagonal pyramid is shown, with units in centimeters (cm). The slant height is 11 cm, the apothem of the base is 4 cm, and the side length of the base is 6 cm.



Which expression represents the surface area, in square centimeters, of the pyramid?

- Ⓐ $\frac{6 \cdot 11 \cdot 5}{2}$
- Ⓑ $6\left(4 + \frac{5 \cdot 11}{2}\right)$
- Ⓒ $\frac{4 \cdot 11}{2}(6 + 5)$
- Ⓓ $\frac{6 \cdot 5}{2}(4 + 11)$

Geometry EOC

- 29.** Ashleigh draws a map of her town on a coordinate grid, where each unit represents 1 mile. A pizza shop in the town delivers to any location within a 6-mile radius.

- Ashleigh's house is located at point (9, 3).
- The pizza shop is located at point (6, 7).

Complete the sentences to describe whether Ashleigh's house is within the delivery radius of the pizza shop.

The shortest distance from the pizza shop to Ashleigh's house is

miles.

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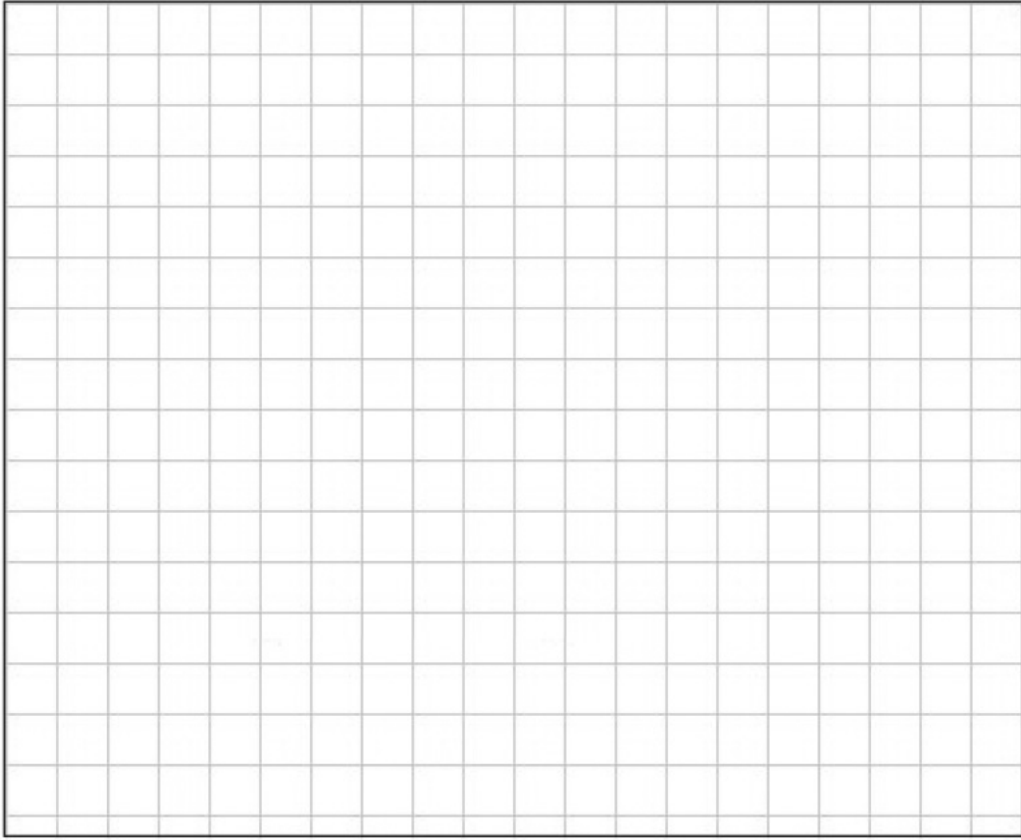
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1	2	3
4	5	6
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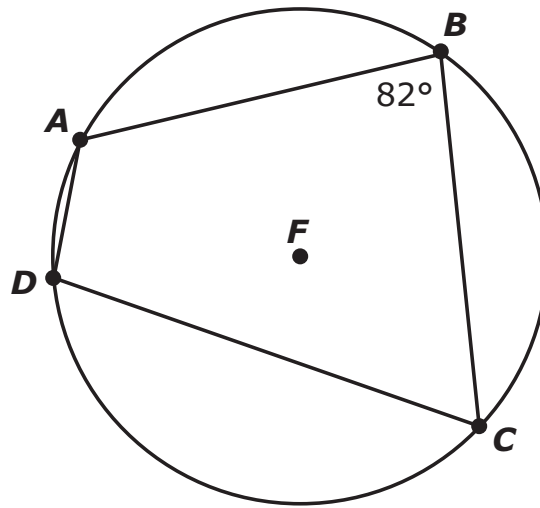
Therefore, Ashleigh's house is [**A** within **B** outside] the delivery radius of the pizza shop.

- 30.** Kyla claims that the sum of the interior angles of any closed figure is 360 degrees or greater.

Use the Connect Line tool to draw a figure that is a counterexample to Kyla's claim.



31. Quadrilateral $ABCD$ is inscribed in circle F , as shown.



Complete the statements to describe the measure, in degrees, of $\angle ADC$.

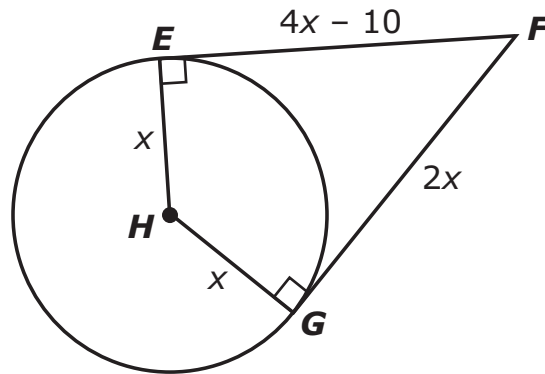
Angle ABC and angle ADC are [Ⓐ complementary. Ⓑ congruent.
Ⓒ supplementary.]

This means the measure of $\angle ADC =$

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1	2	3
4	5	6
7	8	9
	0	
.	-	÷

32. A circle with two tangent segments is shown.



This question has **two** parts.

Part A

Which equation is true?

- Ⓐ $(4x - 10) = 2x$
- Ⓑ $x(4x - 10) = (2x)$
- Ⓒ $(4x - 10) + 2x = 180$
- Ⓓ $x + x + 2x + (4x - 10) = 360$

Part B

What is the value of x ?

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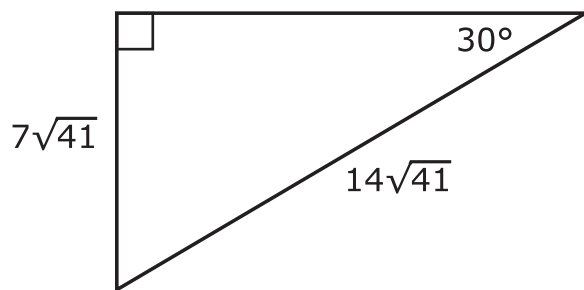
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1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\Box}{\Box}$

33. A right triangle is shown, with side lengths in units.

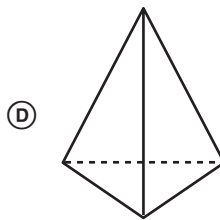
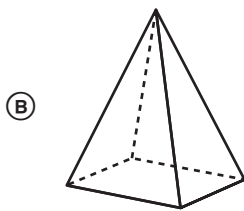
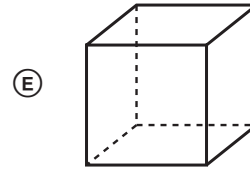
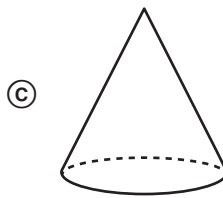
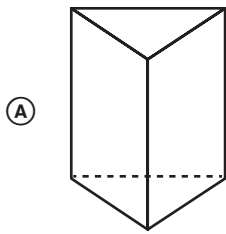


Select an expression to complete the trigonometric ratio for the triangle.

$$\tan(30^\circ) = \frac{7\sqrt{41}}{\quad}$$

[Ⓐ $7\sqrt{41}$ Ⓑ $14\sqrt{41}$ Ⓒ $7\sqrt{123}$ Ⓓ $14\sqrt{123}$]

- 34.** Select all the figures that have a triangular cross section when sliced parallel to their bases.



- 35.** Prism A is dilated by a scale factor greater than 0 and less than 1 to create prism B .

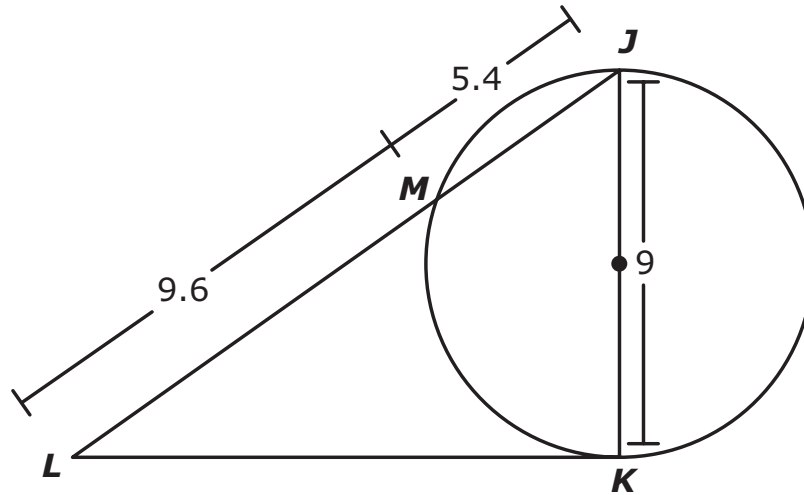
Complete the sentences to compare the volume and surface area of the two prisms.

The volume of prism A is [$\textcircled{A}$ equal to $\textcircled{B}$ less than $\textcircled{C}$ greater than] the volume of prism B .

The surface area of prism A is [$\textcircled{A}$ equal to $\textcircled{B}$ less than $\textcircled{C}$ greater than] the surface area of prism B .

Geometry EOC

- 36.** Line segments JK , KL , and LJ form triangle JKL , as shown, with dimensions in units.



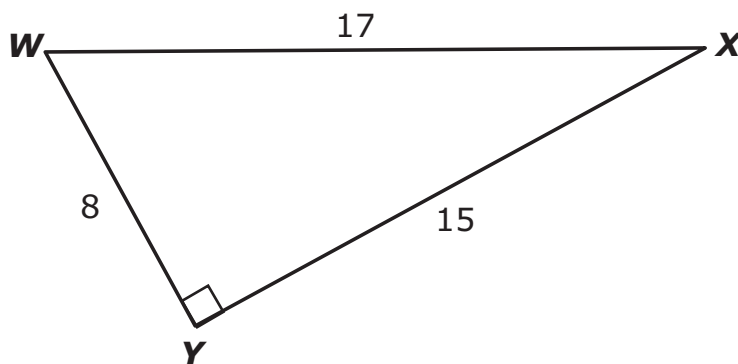
- Segment KL is tangent to the circle.
- Segment LJ is a secant of the circle.
- Segment JK is a diameter of the circle.

What is the perimeter, in units, of triangle JKL ?

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1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\Box}{\Box}$

37. Triangle WXY is shown, with side lengths in units.



Match each trigonometric ratio to its value.

	$\frac{15}{17}$	$\frac{17}{15}$	$\frac{8}{17}$	$\frac{17}{8}$	$\frac{8}{15}$	$\frac{15}{8}$
$\cos (x)$	(A)	(B)	(C)	(D)	(E)	(F)
$\tan (W)$	(G)	(H)	(I)	(J)	(K)	(L)

Geometry EOC

- 38.** Maria has a circular pizza with a diameter of 12 inches. She cuts the pizza so that each slice has a central angle that measures 90 degrees.

What is the area, in square inches, of each slice of pizza?

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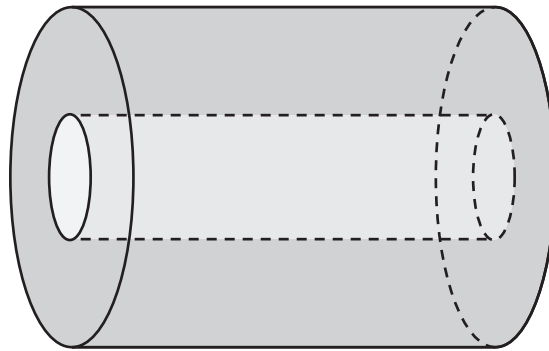
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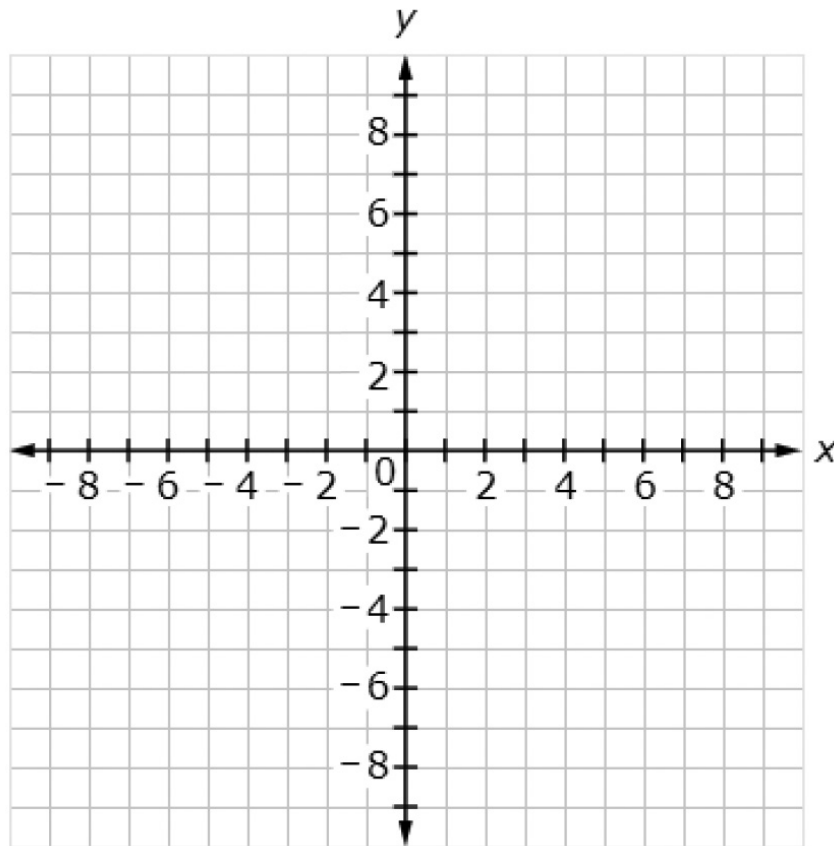
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1	2	3	+	-	•	÷				
4	5	6	<	≤	=	≥	>			
7	8	9	$\square^\square$	$\square_\square$	()		$\sqrt{\square}$	$\sqrt[\square]{\square}$	π	i
	0		sin	cos	tan	arcsin	arccos	arctan		
.	-	$\frac{\square}{\square}$								

- 39.** Karina draws a figure on the coordinate plane. Then, she rotates the figure around the x -axis. The resulting object is shown.



Use the Connect Line tool to create a possible figure that Karina could have drawn.



Geometry EOC

40. The transformations given are performed on pentagon $ABCDE$ to create pentagon $A'B'C'D'E'$.

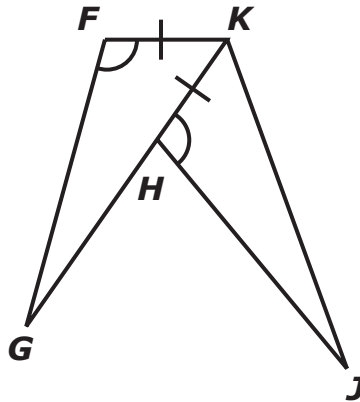
- $(x, y) \rightarrow (x - 5, y - 6)$
- counterclockwise rotation of 37° about the origin

Complete the sentence to explain whether or not the pentagons are congruent.

Pentagons $ABCDE$ and $A'B'C'D'E'$ [Ⓐ are Ⓑ are not] congruent because [Ⓐ each Ⓑ neither Ⓒ only the first Ⓓ only the second] transformation preserves side lengths and angle measures.

41. A figure is shown.

- Segment KG bisects angle FKJ .
- Angles GFK and JHK are congruent.
- Line segments FK and HK are congruent.



Geometry EOC

A partial proof is shown.

Statement	Reason
1. $\overline{FK} \cong \overline{HK}$ and $\angle GFK \cong \angle JHK$	1. Given
2. $\overline{KG}$ bisects $\angle FKJ$	2. Given
3.	3. Definition of angle bisector
4. $\triangle GFK \cong \triangle JHK$	4.

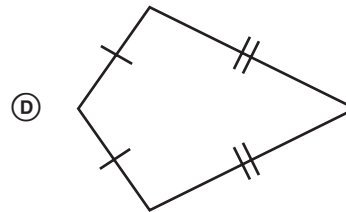
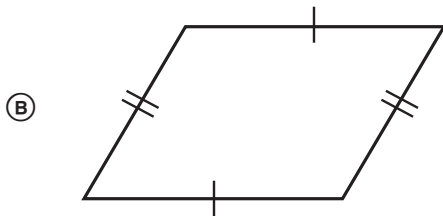
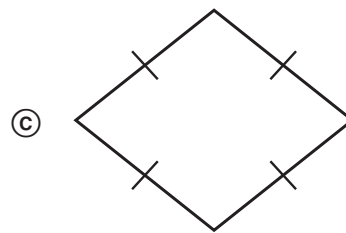
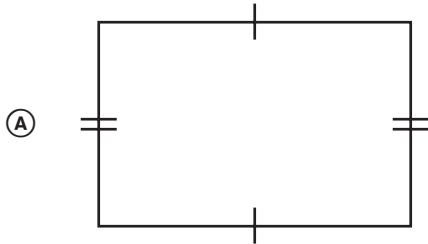
What could be Reason 4?

- Ⓐ Side-Angle-Side congruence
- Ⓑ Hypotenuse-Leg congruence
- Ⓒ Angle-Angle-Side congruence
- Ⓓ Angle-Side-Angle congruence

42. Johanna makes the following claim:

“If a polygon is a quadrilateral, then two pairs of opposite sides are congruent.”

Which polygon is a counterexample to Johanna’s claim?

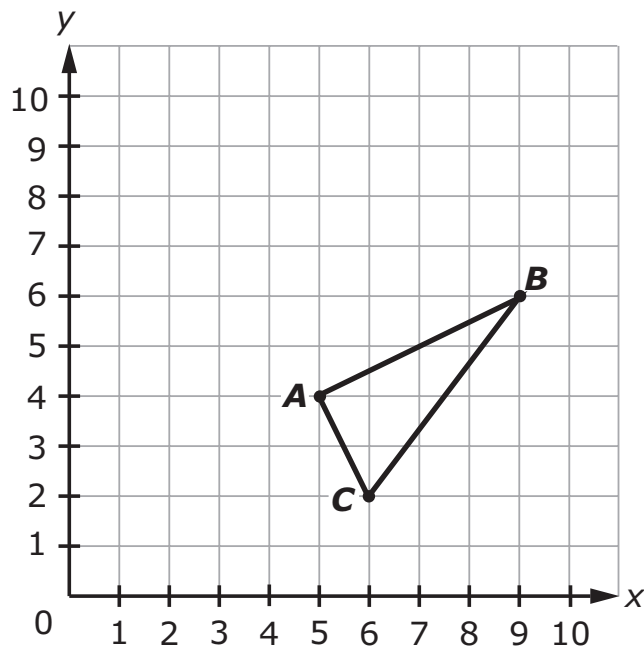


- 43.** A sequence of transformations maps triangle QRS onto congruent triangle XYZ .

Select all the sequences of transformations that could have mapped triangle QRS onto triangle XYZ .

- Ⓐ a reflection across the x -axis, followed by a 90° clockwise rotation about the origin
- Ⓑ a 180° clockwise rotation about the origin, followed by a reflection across the line $y = 3$
- Ⓒ a dilation by a scale factor of 3 centered at the origin, followed by a translation 5 units down
- Ⓓ a reflection across the line $y = x$, followed by a dilation by a scale factor of $\frac{1}{3}$ centered at the origin
- Ⓔ a translation 1 unit left and 3 units up, followed by a 270° counterclockwise rotation about the origin

44. Triangle ABC is shown on the coordinate grid.

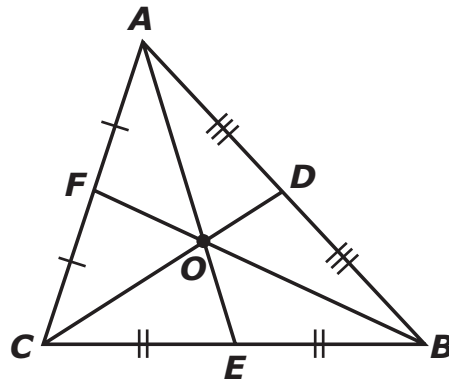


Geometry EOC

Complete the steps to describe a sequence of transformations that will map triangle ABC onto itself.

Step 1	Translate triangle ABC 4 units up.															
Step 2	Translate triangle ABC ← → ↶ ↷ ✕ <table border="1"><tr><td>1</td><td>2</td><td>3</td></tr><tr><td>4</td><td>5</td><td>6</td></tr><tr><td>7</td><td>8</td><td>9</td></tr><tr><td></td><td>0</td><td></td></tr><tr><td>.</td><td>-</td><td>$\frac{\Box}{\Box}$</td></tr></table> units down.	1	2	3	4	5	6	7	8	9		0		.	-	$\frac{\Box}{\Box}$
1	2	3														
4	5	6														
7	8	9														
	0															
.	-	$\frac{\Box}{\Box}$														
Step 3	Dilate triangle ABC by a scale factor of ← → ↶ ↷ ✕ <table border="1"><tr><td>1</td><td>2</td><td>3</td></tr><tr><td>4</td><td>5</td><td>6</td></tr><tr><td>7</td><td>8</td><td>9</td></tr><tr><td></td><td>0</td><td></td></tr><tr><td>.</td><td>-</td><td>$\frac{\Box}{\Box}$</td></tr></table> centered at the origin.	1	2	3	4	5	6	7	8	9		0		.	-	$\frac{\Box}{\Box}$
1	2	3														
4	5	6														
7	8	9														
	0															
.	-	$\frac{\Box}{\Box}$														

45. A figure is shown.



Complete the statement to describe $\triangle ABC$.

The [$\textcircled{A}$ medians $\textcircled{B}$ angle bisectors $\textcircled{C}$ perpendicular bisectors] of $\triangle ABC$ meet at point [$\textcircled{A}$ C . $\textcircled{B}$ D . $\textcircled{C}$ O .]



THE B.E.S.T. STANDARDS

Benchmarks for Excellent Student Thinking

Geometry EOC Assessment

Test Release: 2025

1. Two statements about triangle JKL are shown.

- $m \angle L = 90^\circ$

- $\sin(J) = \frac{99}{101}$

Which ratio represents $\cos(K)$?

Ⓐ $\frac{101}{99}$

Ⓑ $\frac{99}{101}$

Ⓒ $\frac{20}{101}$

Ⓓ $\frac{20}{99}$

Answer Key: B

Percentage of Students Answering Correctly: 40%

Reporting Category: Measurement and Coordinate Geometry

Benchmark: MA.912.T.1.1

Benchmark Description: Define trigonometric ratios for acute angles in right triangles.

Geometry EOC Test Release Support Document

2. Line segment MN has endpoints $M (-7, 10)$ and $N (-1, 1)$. Point P is on line segment MN such that the ratio $MP:PN$ is 2:1.

What is the x -coordinate of point P ?

-3

←

→

↶

↷

✖

1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\Box}{\Box}$

Answer Key: -3, or any equivalent value

Percentage of Students Answering Correctly: 23%

Reporting Category: Measurement and Coordinate Geometry

Benchmark: MA.912.GR.3.3

Benchmark Description: Use coordinate geometry to solve mathematical and real-world geometric problems involving lines, circles, triangles and quadrilaterals.

3. Which statement is true when a three-dimensional figure is dilated by a given scale factor?
- Ⓐ A dilation by a scale factor of $\frac{2}{3}$ results in an increase in volume.
 - Ⓑ A dilation by a scale factor of 0.1 results in no change in volume.
 - Ⓒ A dilation by a scale factor of $\frac{3}{2}$ results in a decrease in surface area.
 - Ⓓ A dilation by a scale factor of 3.5 results in an increase in surface area.

Answer Key: D

Percentage of Students Answering Correctly: 49%

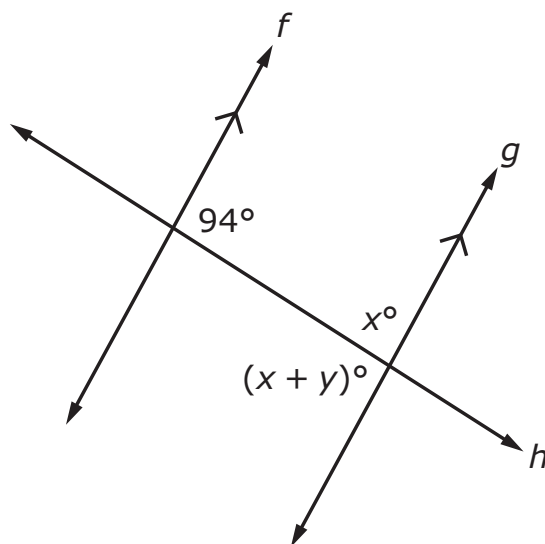
Reporting Category: Congruence, Similarity, and Constructions

Benchmark: MA.912.GR.4.3

Benchmark Description: Extend previous understanding of scale drawings and scale factors to determine how dilations affect the area of two-dimensional figures and the surface area or volume of three-dimensional figures.

Geometry EOC Test Release Support Document

4. Two parallel lines, f and g , are cut by a transversal, h , as shown.



What are the values of x and y ?

$x =$

86

←	→	↶	↷	✖
1	2	3		
4	5	6		
7	8	9		
	0			
.	-	$\frac{\Box}{\Box}$		

Geometry EOC Test Release Support Document

$y =$

8				
←	→	↶	↷	✖
1	2	3		
4	5	6		
7	8	9		
	0			
.	-	$\frac{\square}{\square}$		

Answer Key: 86, 8, or any equivalent values

Percentage of Students Answering Correctly: 34%

Reporting Category: Logic, Relationships, and Theorems

Benchmark: MA.912.GR.1.1

Benchmark Description: Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships and theorems of lines and angles.

5. Which statement must be true for all parallelograms?

- Ⓐ Diagonals are congruent.
- Ⓑ Diagonals are perpendicular.
- Ⓒ Opposite angles are complementary.
- Ⓓ Consecutive angles are supplementary.

Answer Key: D

Percentage of Students Answering Correctly: 29%

Reporting Category: Logic, Relationships, and Theorems

Benchmark: MA.912.GR.1.4

Benchmark Description: Prove relationships and theorems about parallelograms. Solve mathematical and real-world problems involving postulates, relationships and theorems of parallelograms.

6. A true statement is given.

“If two given shapes are circles, then the shapes are similar.”

Select the boxes to identify what could be the converse, inverse, and contrapositive of the given statement.

	Converse	Inverse	Contrapositive
If two given shapes are not similar, then the shapes are not circles.	(A)	(B)	(C)
If two given shapes are similar, then the shapes are circles.	(D)	(E)	(F)
If two given shapes are not circles, then the shapes are not similar.	(G)	(H)	(I)

Answer Key: C, D, H

Percentage of Students Answering Correctly: 28%

Reporting Category: Logic, Relationships, and Theorems

Benchmark: MA.912.LT.4.3

Benchmark Description: Identify and accurately interpret “if...then,” “if and only if,” “all” and “not” statements. Find the converse, inverse and contrapositive of a statement.

Geometry EOC Test Release Support Document

7. This question has **two** parts.

The vertices of two triangles are given.

- The coordinates of the vertices of the first triangle are $Q(2, 0)$, $R(6, 0)$, and $S(2, 4)$.
- The coordinates of the vertices of the second triangle are $T(8, 0)$, $U(4, -4)$, and $V(4, 0)$.

Part A

Which statement describes a relationship between the first and second triangles?

- ☐ Ⓐ $\triangle QRS \cong \triangle TUV$
- ☒ Ⓑ $\triangle QRS \cong \triangle VTU$
- ☐ Ⓒ $\triangle QRS \sim \triangle TUV$
- ☐ Ⓓ $\triangle QRS \sim \triangle UVT$

Part B

Select expressions to create a transformation that proves the statement in Part A.

$$(x, y) \rightarrow ([\textcircled{A} 2x \textcircled{B} -x \textcircled{C} x + 2 \textcircled{D} x - 2], [\textcircled{A} 2y \textcircled{B} -y \textcircled{C} y + 8 \textcircled{D} y - 8])$$

Answer Key: B, C, B

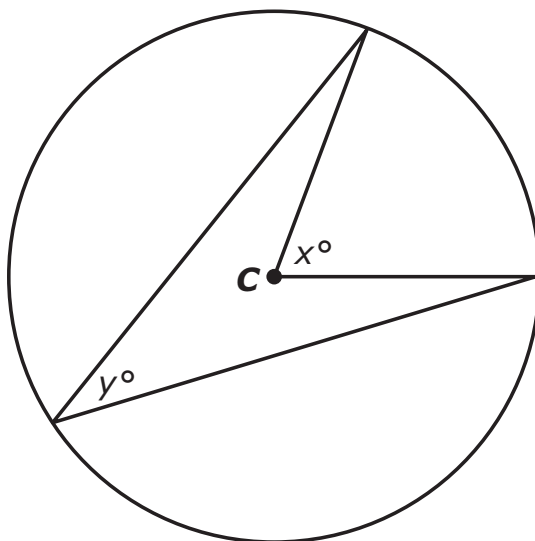
Percentage of Students Answering Correctly: 21%

Reporting Category: Congruence, Similarity, and Constructions

Benchmark: MA.912.GR.2.6

Benchmark Description: Apply rigid transformations to map one figure onto another to justify that the two figures are congruent.

- 8.** An inscribed angle and a central angle are shown in circle C .



Create an equation to show the relationship between x and y .

$$x = 2y$$

← → ↶ ↷ ✕														
1	2	3	x	y										
4	5	6	+	-	•	÷								
7	8	9	<	≤	=	≥	>							
	0		$\square^{\square}$	$\square_{\square}$	()		$\sqrt{\square}$	$\sqrt[\square]{\square}$	π	i				
.	-	$\frac{\square}{\square}$	sin	cos	tan	arcsin	arccos	arctan						

Answer Key: $x = 2y$, or any equivalent equation

Percentage of Students Answering Correctly: 25%

Reporting Category: Logic, Relationships, and Theorems

Benchmark: MA.912.GR.6.2

Benchmark Description: Solve mathematical and real-world problems involving the measures of arcs and related angles.

Geometry EOC Test Release Support Document

9. Figure A is drawn on a coordinate plane. Transformations are performed on figure A to create figure B.

Select the boxes to identify whether each transformation performed on figure A results in a congruent or noncongruent figure B.

	Congruent	Noncongruent
$(x, y) \rightarrow (y, -x)$	☒ A	☐ B
$(x, y) \rightarrow \left(\frac{1}{2}x, \frac{1}{2}y\right)$	☐ C	☒ D
a reflection across the line $y = -x$	☒ E	☐ F
a 50-degree clockwise rotation about the origin	☒ G	☐ H

Answer Key: A, D, E, G

Percentage of Students Answering Correctly: 30%

Reporting Category: Congruence, Similarity, and Constructions

Benchmark: MA.912.GR.2.2

Benchmark Description: Identify transformations that do or do not preserve distance.

10. A line segment has an endpoint at $(1, 1)$ and a midpoint at $(1, -3)$.

What are the coordinates of the other endpoint?

- Ⓐ $(1, 5)$
- Ⓑ $(1, -1)$
- Ⓒ $(1, -5)$
- Ⓓ $(1, -7)$

Answer Key: D

Percentage of Students Answering Correctly: 50%

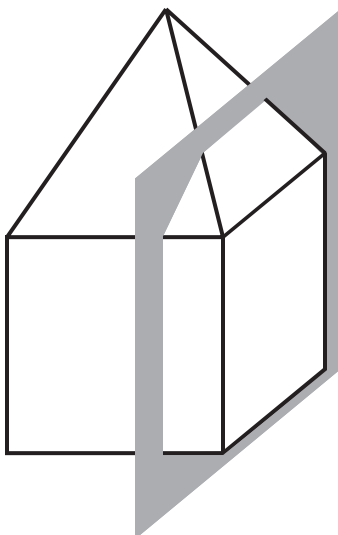
Reporting Category: Measurement and Coordinate Geometry

Benchmark: MA.912.GR.3.3

Benchmark Description: Use coordinate geometry to solve mathematical and real-world geometric problems involving lines, circles, triangles and quadrilaterals.

Geometry EOC Test Release Support Document

- 11.** The figure consisting of a right pyramid and a right rectangular prism is sliced perpendicular to its base, as shown.



What is the shape of the resulting cross section?

- Ⓐ a rectangle
- Ⓑ a trapezoid
- Ⓒ a composite shape made up of a rectangle and a triangle
- Ⓓ a composite shape made up of a rectangle and a trapezoid

Answer Key: D

Percentage of Students Answering Correctly: 55%

Reporting Category: Measurement and Coordinate Geometry

Benchmark: MA.912.GR.4.1

Benchmark Description: Identify the shapes of two-dimensional cross-sections of three-dimensional figures.

12. A triangle has vertices at $T(2, 5)$, $U(6, 6)$, and $V(4, -3)$.

Complete the statements to explain why triangle TUV is a right triangle.

The slope of $\overline{TU}$ is

←
→
↶
↷
✖

1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\Box}{\Box}$

and the slope of $\overline{TV}$ is

←
→
↶
↷
✖

1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\Box}{\Box}$

Triangle TUV is a right triangle because $\overline{TU}$ and $\overline{TV}$ are
 [Ⓐ skewed Ⓑ parallel Ⓒ perpendicular] and therefore form a right angle.

Geometry EOC Test Release Support Document

Answer Key: $\frac{1}{4}$, -4 (or any equivalent values), C

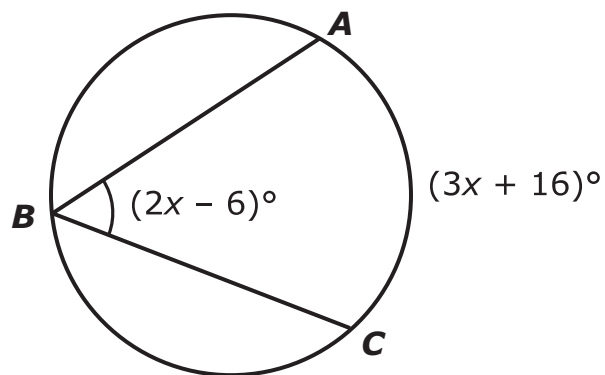
Percentage of Students Answering Correctly: 24%

Reporting Category: Measurement and Coordinate Geometry

Benchmark: MA.912.GR.3.2

Benchmark Description: Given a mathematical or real-world context, use coordinate geometry to classify or justify definitions, properties and theorems involving circles, triangles or quadrilaterals.

13. Points A , B , and C lie on the circle, as shown.



- The measure of $\angle ABC$ is $(2x - 6)^\circ$.
- The measure of $\widehat{AC}$ is $(3x + 16)^\circ$.

What is the measure, in degrees, of $\angle ABC$?

50

←
→
↶
↷
✖

1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\Box}{\Box}$

Answer Key: 50, or any equivalent value

Percentage of Students Answering Correctly: 21%

Reporting Category: Logic, Relationships, and Theorems

Benchmark: MA.912.GR.6.2

Benchmark Description: Solve mathematical and real-world problems involving the measures of arcs and related angles.

Geometry EOC Test Release Support Document

- 14.** A cone has a radius of 3 units, a height of 4 units, and a slant height of 5 units.

Which expression represents the surface area, in square units, of the cone?

- Ⓐ 15π
- Ⓑ 21π
- Ⓒ 24π
- Ⓓ 29π

Answer Key: C

Percentage of Students Answering Correctly: 43%

Reporting Category: Measurement and Coordinate Geometry

Benchmark: MA.912.GR.4.6

Benchmark Description: Solve mathematical and real-world problems involving the surface area of three-dimensional figures limited to cylinders, pyramids, prisms, cones and spheres.

15. Rectangle $ABCD$ has the given side lengths.

- $AB = 5$ units
- $BC = 11$ units

Rectangle $PQRS$ is congruent to rectangle $ABCD$.

What is the length of $\overline{QR}$, in units?

11

←
→
↶
↷
✖

1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\Box}{\Box}$

Answer Key: 11, or any equivalent value

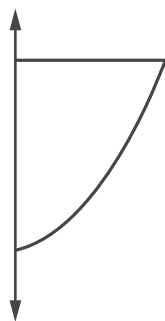
Percentage of Students Answering Correctly: 63%

Reporting Category: Congruence, Similarity, and Constructions

Benchmark: MA.912.GR.1.6

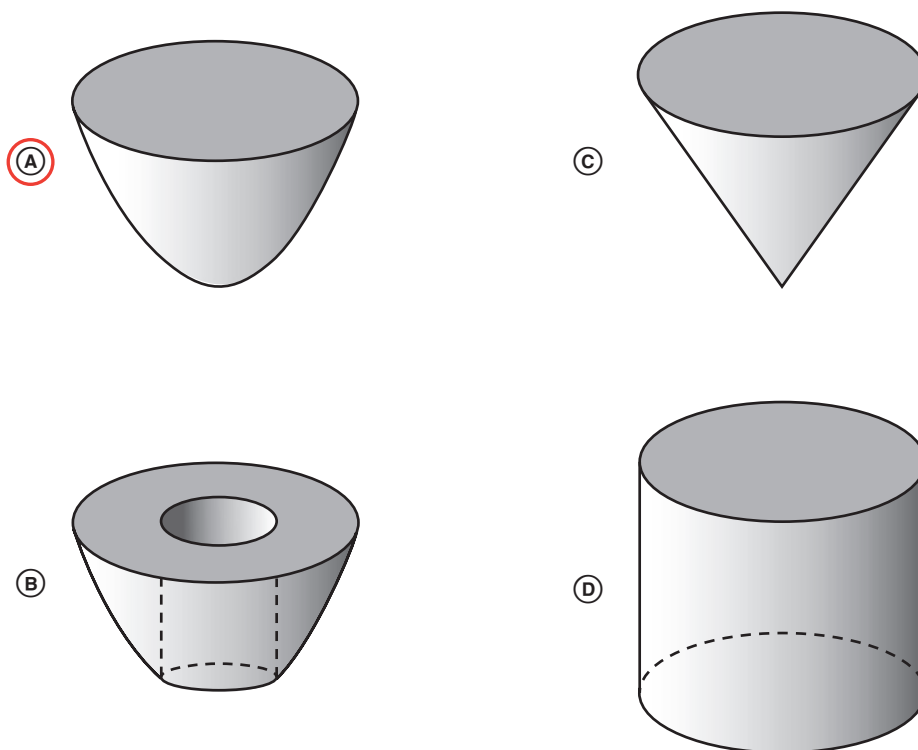
Benchmark Description: Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures.

16. A two-dimensional figure and a vertical line are shown.



The figure is rotated about the vertical line.

Which object could be the result of this rotation?



Answer Key: A

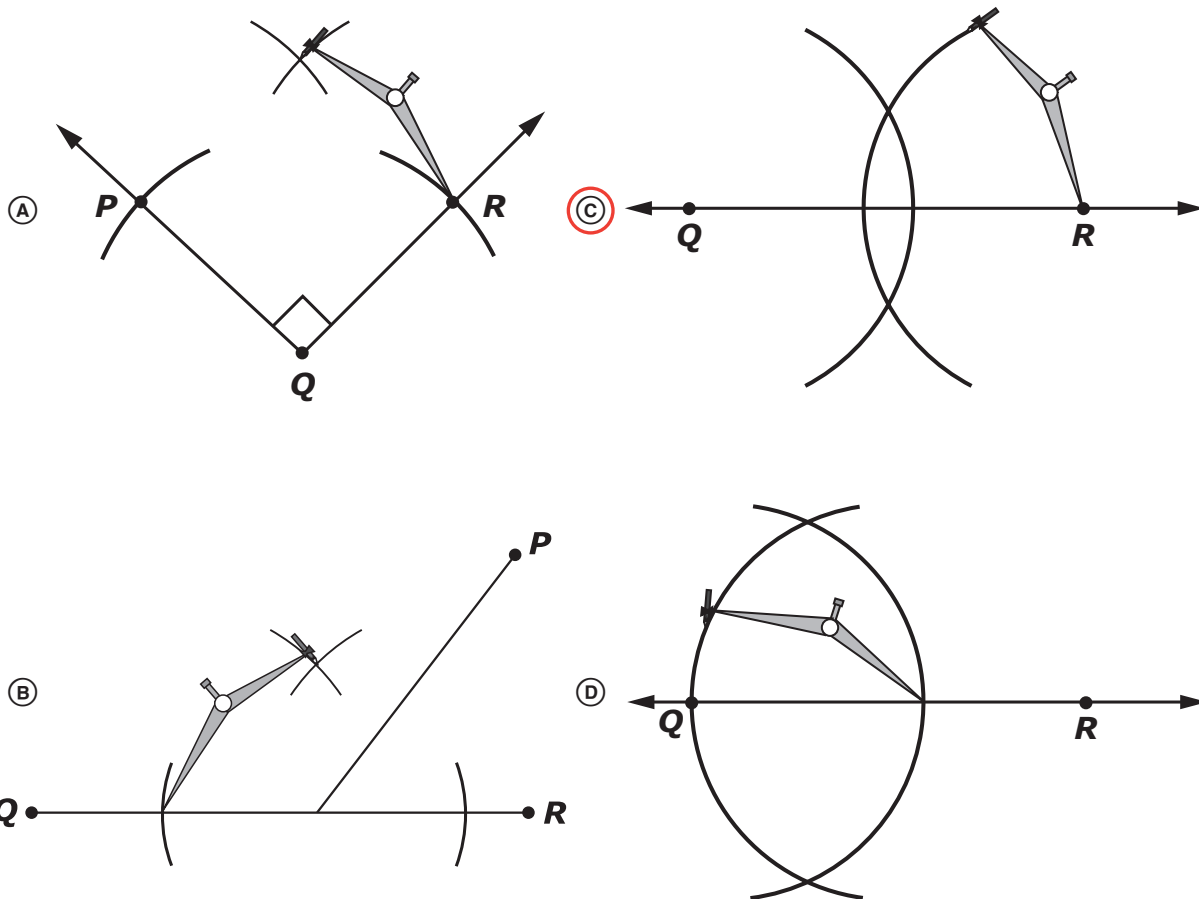
Percentage of Students Answering Correctly: 81%

Reporting Category: Measurement and Coordinate Geometry

Benchmark: MA.912.GR.4.2

Benchmark Description: Identify three-dimensional objects generated by rotations of two-dimensional figures.

17. Which figure shows steps leading to the construction of a perpendicular bisector?



Answer Key: C

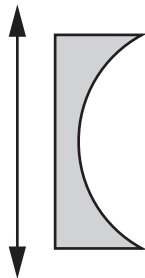
Percentage of Students Answering Correctly: 39%

Reporting Category: Congruence, Similarity, and Constructions

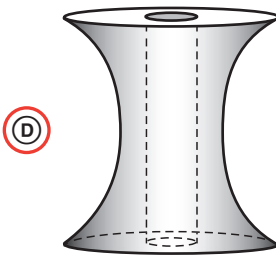
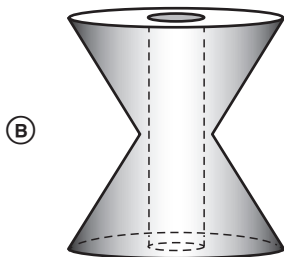
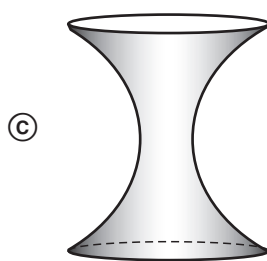
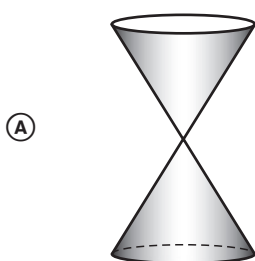
Benchmark: MA.912.GR.5.2

Benchmark Description: Construct the bisector of a segment or an angle, including the perpendicular bisector of a line segment.

18. A figure and a vertical line are shown.



Which object is generated by rotating the figure around the vertical line?



Answer Key: D

Percentage of Students Answering Correctly: 60%

Reporting Category: Measurement and Coordinate Geometry

Benchmark: MA.912.GR.4.2

Benchmark Description: Identify three-dimensional objects generated by rotations of two-dimensional figures.

- 19.** Triangle M has a base length of 18 inches and a height of 12 inches. Triangle N is congruent to triangle M .

What is the area, in square inches, of triangle N ?

108

← → ↶ ↷ ✖

1	2	3	
4	5	6	
7	8	9	
	0		
.	-	$\frac{\Box}{\Box}$	

Answer Key: 108, or any equivalent value

Percentage of Students Answering Correctly: 34%

Reporting Category: Congruence, Similarity, and Constructions

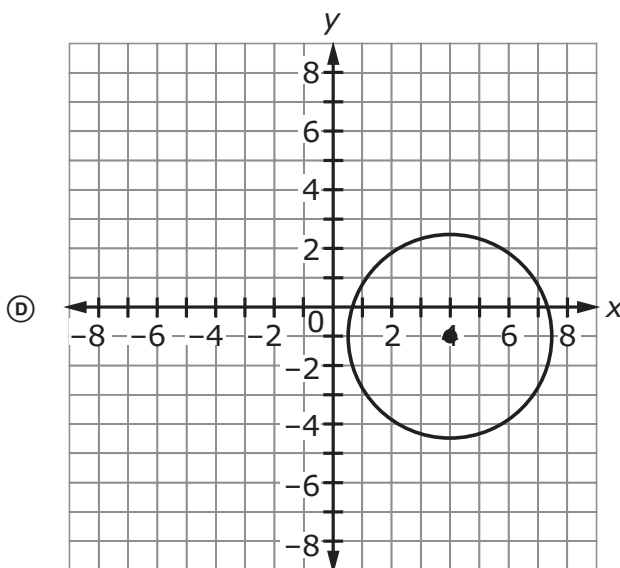
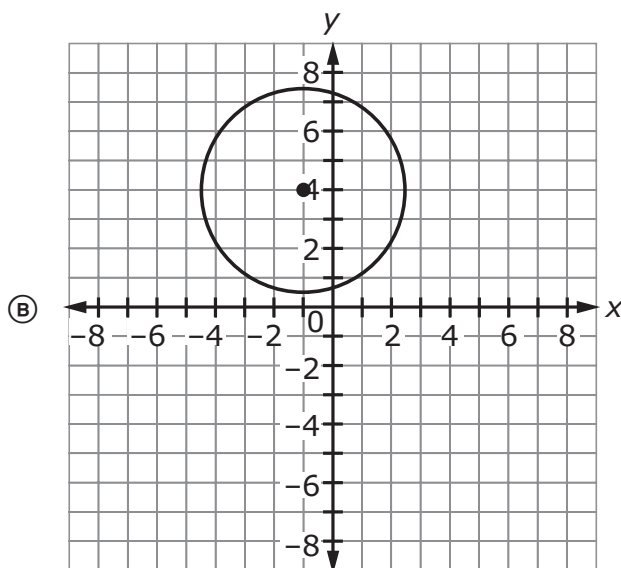
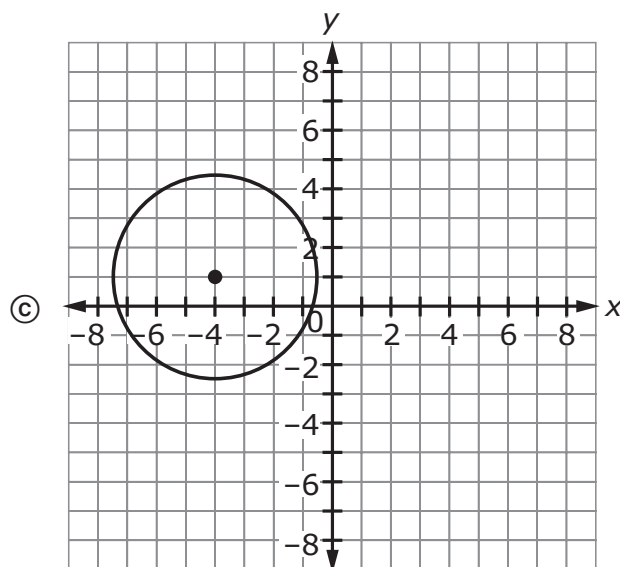
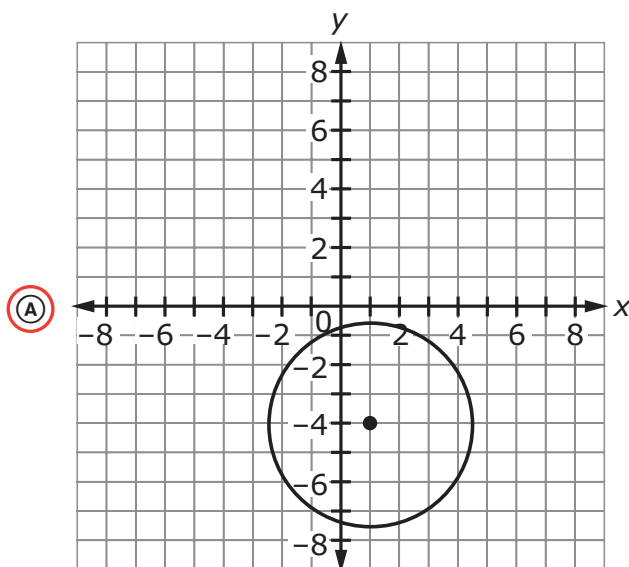
Benchmark: MA.912.GR.1.6

Benchmark Description: Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures.

20. The equation of a circle is given.

$$(x - 1)^2 + (y + 4)^2 = 12.25$$

Which graph represents the equation?



Geometry EOC Test Release Support Document

Answer Key: A

Percentage of Students Answering Correctly: 34%

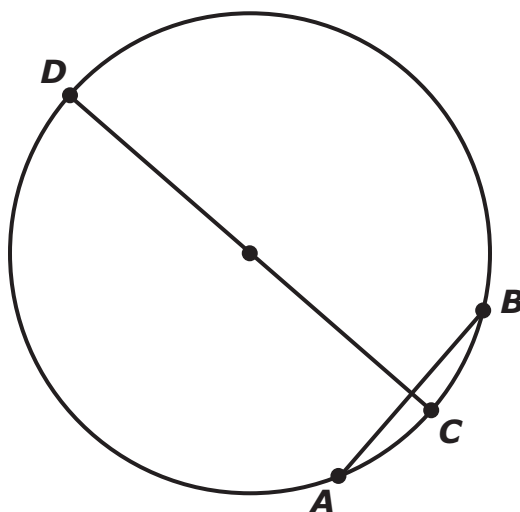
Reporting Category: Measurement and Coordinate Geometry

Benchmark: MA.912.GR.7.3

Benchmark Description: Graph and solve mathematical and real-world problems that are modeled with an equation of a circle. Determine and interpret key features in terms of the context.

Geometry EOC Test Release Support Document

21. A circle with chord AB and its perpendicular bisector CD is shown.



The measure of $\widehat{AC}$ is 28° .

What is the measure, in degrees, of $\widehat{BD}$?

152

←

→

↶

↷

✖

1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\Box}{\Box}$

Answer Key: 152, or any equivalent value

Percentage of Students Answering Correctly: 42%

Reporting Category: Logic, Relationships, and Theorems

Benchmark: MA.912.GR.6.2

Benchmark Description: Solve mathematical and real-world problems involving the measures of arcs and related angles.

22. Triangle XYZ is transformed to create triangle $X'Y'Z'$ using the rule $(x, y) \rightarrow (x + b, y + c)$.

- Triangle XYZ has the vertices $(3, -2)$, $(3, -4)$, and $(0, -4)$, respectively.
- Triangle $X'Y'Z'$ has the vertices $(-1, 5)$, $(-1, 3)$, and $(-4, 3)$, respectively.

What are the values of b and c ?

$b =$

1	2	3		
4	5	6		
7	8	9		
	0			
.	-	$\frac{\Box}{\Box}$		

$c =$

1	2	3		
4	5	6		
7	8	9		
	0			
.	-	$\frac{\Box}{\Box}$		

Geometry EOC Test Release Support Document

Answer Key: -4 , 7 , or any equivalent values

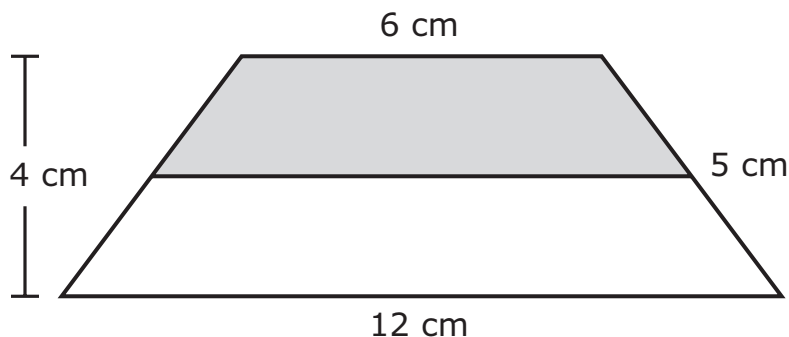
Percentage of Students Answering Correctly: 46%

Reporting Category: Congruence, Similarity, and Constructions

Benchmark: MA.912.GR.2.1

Benchmark Description: Given a preimage and image, describe the transformation and represent the transformation algebraically using coordinates.

- 23.** A tile is in the shape of an isosceles trapezoid. The tile is divided into 2 parts by its midsegment. The trapezoid, with lengths in centimeters (cm), is shown.



The top part is shaded.

What is the area, in square centimeters, of the shaded part?

- Ⓐ 12
- Ⓑ 15
- Ⓒ 18
- Ⓓ 36

Answer Key: B

Percentage of Students Answering Correctly: 37%

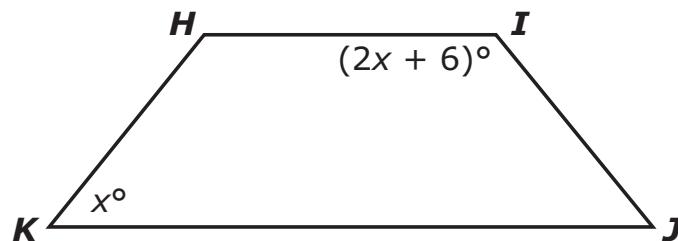
Reporting Category: Logic, Relationships, and Theorems

Benchmark: MA.912.GR.1.5

Benchmark Description: Prove relationships and theorems about trapezoids. Solve mathematical and real-world problems involving postulates, relationships and theorems of trapezoids.

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24. An isosceles trapezoid is shown.



What is the measure of $\angle HIJ$, in degrees?

122

← → ↶ ↷ ✕

1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\Box}{\Box}$

Answer Key: 122, or any equivalent value

Percentage of Students Answering Correctly: 29%

Reporting Category: Logic, Relationships, and Theorems

Benchmark: MA.912.GR.1.5

Benchmark Description: Prove relationships and theorems about trapezoids. Solve mathematical and real-world problems involving postulates, relationships and theorems of trapezoids.

- 25.** Rectangle G has a length of 15 inches and a width of 20 inches. Rectangle H is similar to rectangle G and has a length of 18 feet.

What is the width, in feet, of rectangle H ?

24

←
→
↶
↷
✖

1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\Box}{\Box}$

Answer Key: 24, or any equivalent value

Percentage of Students Answering Correctly: 26%

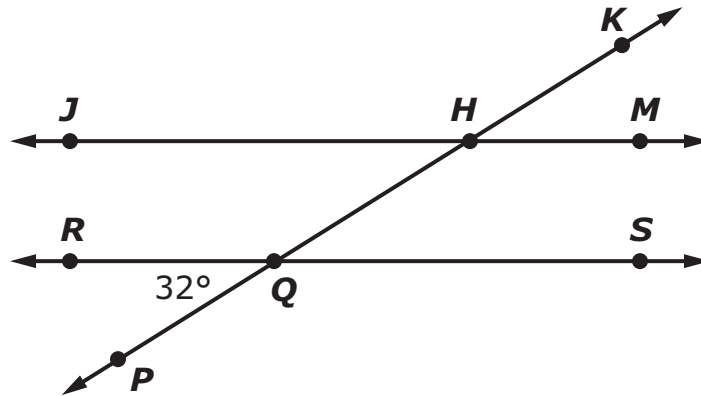
Reporting Category: Congruence, Similarity, and Constructions

Benchmark: MA.912.GR.1.6

Benchmark Description: Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures.

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26. Transversal line KP passes through parallel lines JM and RS , as shown.



Complete the sentences to describe angle QHJ .

Angles PQR and QHJ are [☒ A] congruent [☐ B] supplementary
[☐ C] complementary] because they are [☐ A] consecutive [☒ B] corresponding
[☐ C] alternate exterior] angles.

The measure of angle QHJ is

32

←

→

↶

↷

✖

1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\Box}{\Box}$

Answer Key: A, B, 32 (or any equivalent value)

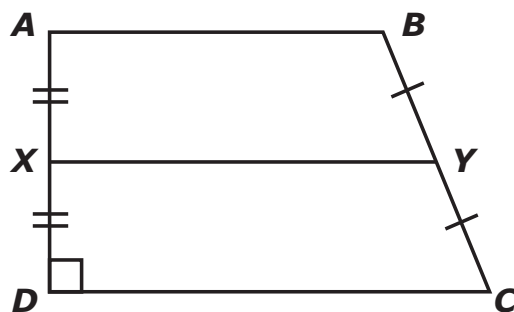
Percentage of Students Answering Correctly: 38%

Reporting Category: Logic, Relationships, and Theorems

Benchmark: MA.912.GR.1.1

Benchmark Description: Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships and theorems of lines and angles.

27. Trapezoid $ABCD$ is shown.



Select all the true statements.

- Ⓐ $\overline{AX} \cong \overline{BY}$
- Ⓑ $\overline{DX} \cong \overline{CY}$
- Ⓒ $\overline{XY} \parallel \overline{CD}$
- Ⓓ $XY = \frac{1}{2}(AB + CD)$
- Ⓔ trapezoid $ABYX \cong$ trapezoid $XYCD$

Answer Key: C, D

Percentage of Students Answering Correctly: 23%

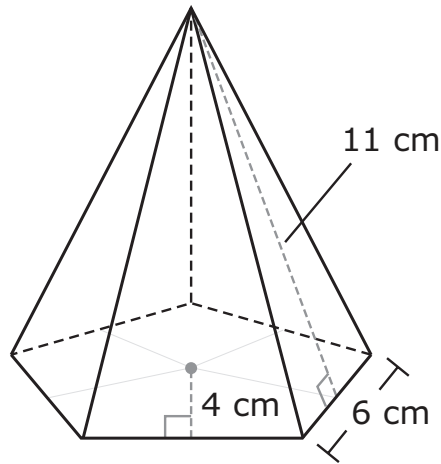
Reporting Category: Logic, Relationships, and Theorems

Benchmark: MA.912.GR.1.5

Benchmark Description: Prove relationships and theorems about trapezoids. Solve mathematical and real-world problems involving postulates, relationships and theorems of trapezoids.

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- 28.** A regular pentagonal pyramid is shown, with units in centimeters (cm). The slant height is 11 cm, the apothem of the base is 4 cm, and the side length of the base is 6 cm.



Which expression represents the surface area, in square centimeters, of the pyramid?

- Ⓐ $\frac{6 \cdot 11 \cdot 5}{2}$
- Ⓑ $6\left(4 + \frac{5 \cdot 11}{2}\right)$
- Ⓒ $\frac{4 \cdot 11}{2}(6 + 5)$
- Ⓓ $\frac{6 \cdot 5}{2}(4 + 11)$

Answer Key: D

Percentage of Students Answering Correctly: 22%

Reporting Category: Measurement and Coordinate Geometry

Benchmark: MA.912.GR.4.6

Benchmark Description: Solve mathematical and real-world problems involving the surface area of three-dimensional figures limited to cylinders, pyramids, prisms, cones and spheres.

29. Ashleigh draws a map of her town on a coordinate grid, where each unit represents 1 mile. A pizza shop in the town delivers to any location within a 6-mile radius.

- Ashleigh's house is located at point (9, 3).
- The pizza shop is located at point (6, 7).

Complete the sentences to describe whether Ashleigh's house is within the delivery radius of the pizza shop.

The shortest distance from the pizza shop to Ashleigh's house is

5

miles.

←
→
↶
↷
✕

1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\Box}{\Box}$

Therefore, Ashleigh's house is [(A) within (B) outside] the delivery radius of the pizza shop.

Answer Key: 5 (or any equivalent value), A

Percentage of Students Answering Correctly: 34%

Reporting Category: Measurement and Coordinate Geometry

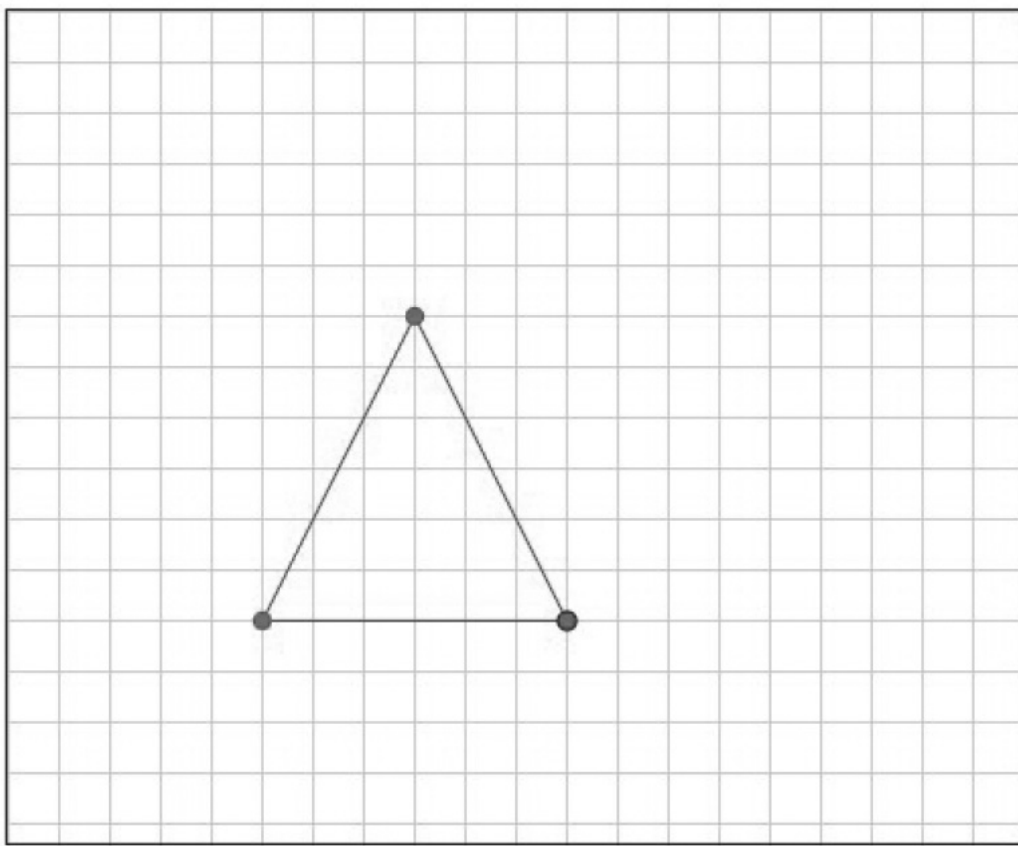
Benchmark: MA.912.GR.3.2

Benchmark Description: Given a mathematical or real-world context, use coordinate geometry to classify or justify definitions, properties and theorems involving circles, triangles or quadrilaterals.

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- 30.** Kyla claims that the sum of the interior angles of any closed figure is 360 degrees or greater.

Use the Connect Line tool to draw a figure that is a counterexample to Kyla's claim.



Answer Key: any triangle

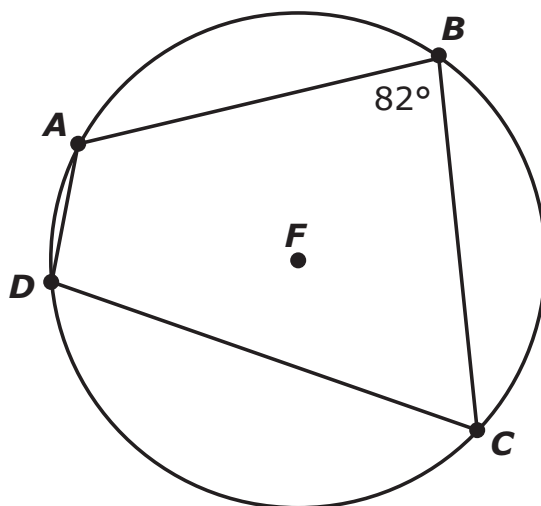
Percentage of Students Answering Correctly: 51%

Reporting Category: Logic, Relationships, and Theorems

Benchmark: MA.912.LT.4.10

Benchmark Description: Judge the validity of arguments and give counterexamples to disprove statements.

31. Quadrilateral $ABCD$ is inscribed in circle F , as shown.



Complete the statements to describe the measure, in degrees, of $\angle ADC$.

Angle ABC and angle ADC are [☐ $\text{\textcircled{A}}$ complementary. ☐ $\text{\textcircled{B}}$ congruent. ☒ $\text{\textcircled{C}}$ supplementary.]

This means the measure of $\angle ADC =$

98

$^\circ$
 $.$

←
→
↶
↷
✕

1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\Box}{\Box}$

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Answer Key: C, 98° (or any equivalent value)

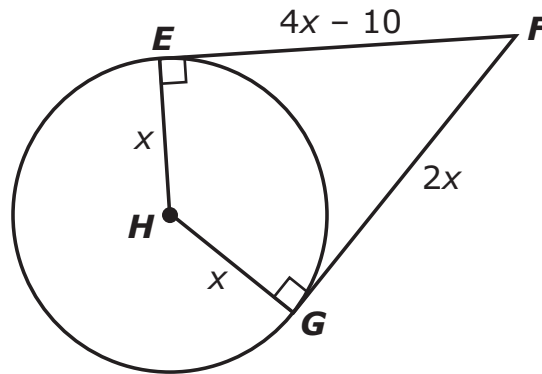
Percentage of Students Answering Correctly: 33%

Reporting Category: Logic, Relationships, and Theorems

Benchmark: MA.912.GR.6.3

Benchmark Description: Solve mathematical problems involving triangles and quadrilaterals inscribed in a circle.

32. A circle with two tangent segments is shown.



This question has **two** parts.

Part A

Which equation is true?

- Ⓐ $(4x - 10) = 2x$
- Ⓑ $x(4x - 10) = (2x)$
- Ⓒ $(4x - 10) + 2x = 180$
- Ⓓ $x + x + 2x + (4x - 10) = 360$

Part B

What is the value of x ?

5

←

→

↶

↷

✖

1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\Box}{\Box}$

Answer Key: A, 5 (or any equivalent value)

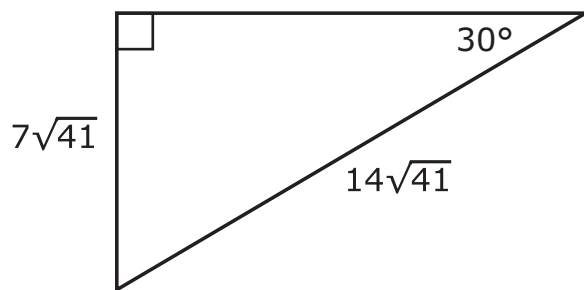
Percentage of Students Answering Correctly: 51%

Reporting Category: Logic, Relationships, and Theorems

Benchmark: MA.912.GR.6.1

Benchmark Description: Solve mathematical and real-world problems involving the length of a secant, tangent, segment or chord in a given circle.

33. A right triangle is shown, with side lengths in units.



Select an expression to complete the trigonometric ratio for the triangle.

$$\tan(30^\circ) = \frac{7\sqrt{41}}{[\textcircled{A} 7\sqrt{41} \quad \textcircled{B} 14\sqrt{41} \quad \textcircled{C} 7\sqrt{123} \quad \textcircled{D} 14\sqrt{123}]}$$

Answer Key: C

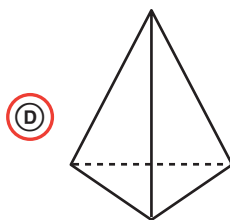
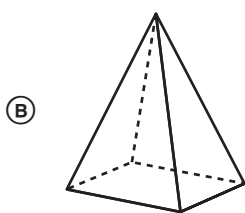
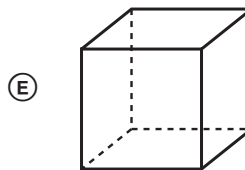
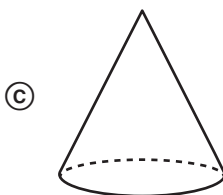
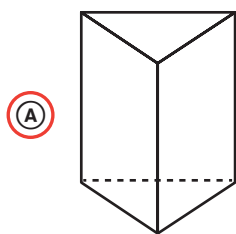
Percentage of Students Answering Correctly: 36%

Reporting Category: Measurement and Coordinate Geometry

Benchmark: MA.912.T.1.2

Benchmark Description: Solve mathematical and real-world problems involving right triangles using trigonometric ratios and the Pythagorean Theorem.

- 34.** Select all the figures that have a triangular cross section when sliced parallel to their bases.



Answer Key: A, D

Percentage of Students Answering Correctly: 44%

Reporting Category: Measurement and Coordinate Geometry

Benchmark: MA.912.GR.4.1

Benchmark Description: Identify the shapes of two-dimensional cross-sections of three-dimensional figures.

- 35.** Prism A is dilated by a scale factor greater than 0 and less than 1 to create prism B .

Complete the sentences to compare the volume and surface area of the two prisms.

The volume of prism A is [☐ A equal to ☐ B less than ☒ C greater than] the volume of prism B .

The surface area of prism A is [☐ A equal to ☐ B less than ☒ C greater than] the surface area of prism B .

Answer Key: C, C

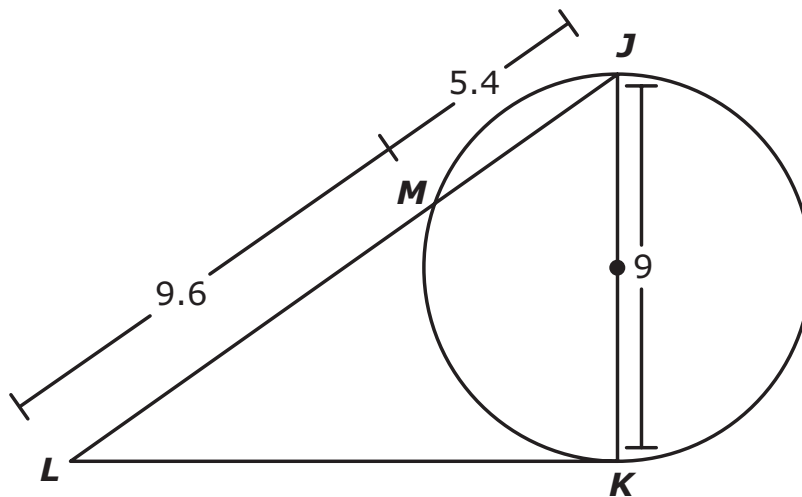
Percentage of Students Answering Correctly: 30%

Reporting Category: Congruence, Similarity, and Constructions

Benchmark: MA.912.GR.4.3

Benchmark Description: Extend previous understanding of scale drawings and scale factors to determine how dilations affect the area of two-dimensional figures and the surface area or volume of three-dimensional figures.

36. Line segments JK , KL , and LJ form triangle JKL , as shown, with dimensions in units.



- Segment KL is tangent to the circle.
- Segment LJ is a secant of the circle.
- Segment JK is a diameter of the circle.

What is the perimeter, in units, of triangle JKL ?

36

←
→
↶
↷
✕

1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\Box}{\Box}$

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Answer Key: 36, or any equivalent value

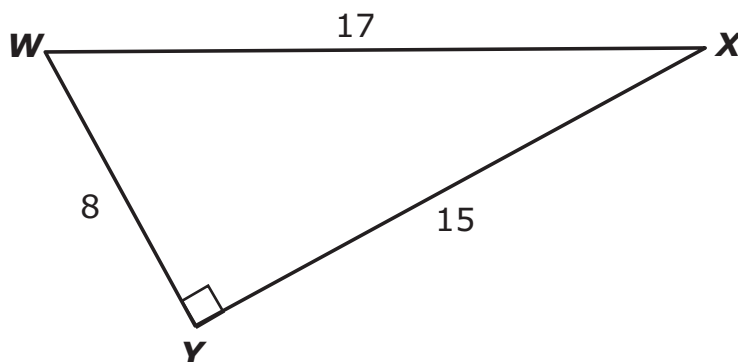
Percentage of Students Answering Correctly: 28%

Reporting Category: Logic, Relationships, and Theorems

Benchmark: MA.912.GR.6.1

Benchmark Description: Solve mathematical and real-world problems involving the length of a secant, tangent, segment or chord in a given circle.

37. Triangle WXY is shown, with side lengths in units.



Match each trigonometric ratio to its value.

	$\frac{15}{17}$	$\frac{17}{15}$	$\frac{8}{17}$	$\frac{17}{8}$	$\frac{8}{15}$	$\frac{15}{8}$
$\cos (x)$	(A)	(B)	(C)	(D)	(E)	(F)
$\tan (W)$	(G)	(H)	(I)	(J)	(K)	(L)

Answer Key: A, L

Percentage of Students Answering Correctly: 31%

Reporting Category: Measurement and Coordinate Geometry

Benchmark: MA.912.T.1.1

Benchmark Description: Define trigonometric ratios for acute angles in right triangles.

- 38.** Maria has a circular pizza with a diameter of 12 inches. She cuts the pizza so that each slice has a central angle that measures 90 degrees.

What is the area, in square inches, of each slice of pizza?

9π

←
→
↶
↷
✕

1	2	3	+	-	•	÷				
4	5	6	<	≤	=	≥	>			
7	8	9	$\square^\square$	$\square_\square$	()		$\sqrt{\square}$	$\sqrt[\square]{\square}$	π	i
	0		sin	cos	tan	arcsin	arccos	arctan		
.	-	$\frac{\square}{\square}$								

Answer Key: 9π , or any value between 28 and 28.3, inclusive

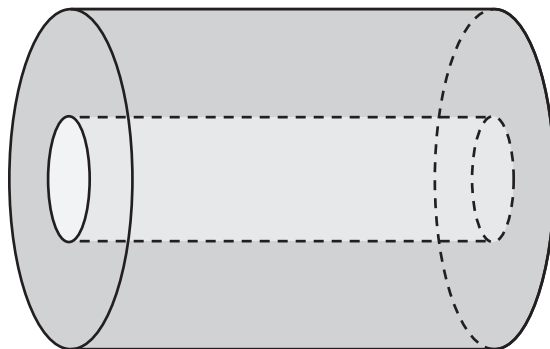
Percentage of Students Answering Correctly: 28%

Reporting Category: Logic, Relationships, and Theorems

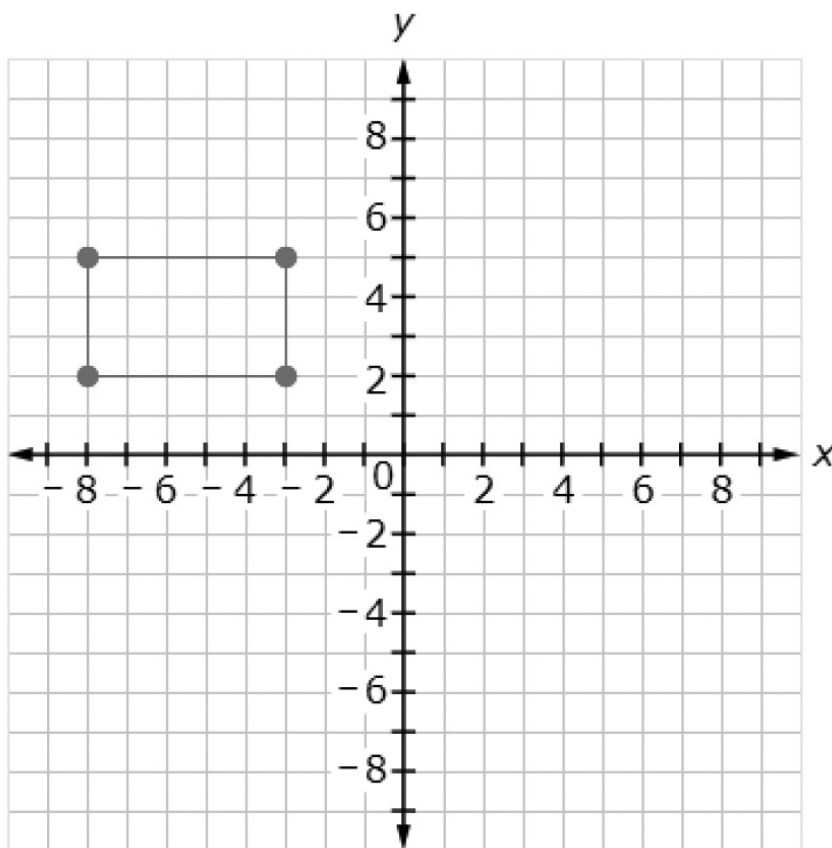
Benchmark: MA.912.GR.6.4

Benchmark Description: Solve mathematical and real-world problems involving the arc length and area of a sector in a given circle.

- 39.** Karina draws a figure on the coordinate plane. Then, she rotates the figure around the x -axis. The resulting object is shown.



Use the Connect Line tool to create a possible figure that Karina could have drawn.



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Answer Key: any rectangle not touching the x-axis

Percentage of Students Answering Correctly: 33%

Reporting Category: Measurement and Coordinate Geometry

Benchmark: MA.912.GR.4.2

Benchmark Description: Identify three-dimensional objects generated by rotations of two-dimensional figures.

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40. The transformations given are performed on pentagon $ABCDE$ to create pentagon $A'B'C'D'E'$.

- $(x, y) \rightarrow (x - 5, y - 6)$
- counterclockwise rotation of 37° about the origin

Complete the sentence to explain whether or not the pentagons are congruent.

Pentagons $ABCDE$ and $A'B'C'D'E'$ ☒ (A) are ☐ (B) are not] congruent because ☒ (A) each ☐ (B) neither ☐ (C) only the first ☐ (D) only the second] transformation preserves side lengths and angle measures.

Answer Key: A, A

Percentage of Students Answering Correctly: 53%

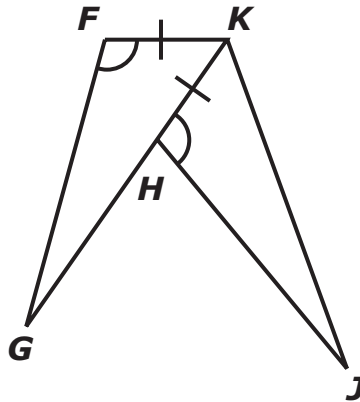
Reporting Category: Congruence, Similarity, and Constructions

Benchmark: MA.912.GR.2.2

Benchmark Description: Identify transformations that do or do not preserve distance.

41. A figure is shown.

- Segment KG bisects angle FKJ .
- Angles GFK and JHK are congruent.
- Line segments FK and HK are congruent.



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A partial proof is shown.

Statement	Reason
1. $\overline{FK} \cong \overline{HK}$ and $\angle GFK \cong \angle JHK$	1. Given
2. $\overline{KG}$ bisects $\angle FKJ$	2. Given
3.	3. Definition of angle bisector
4. $\triangle GFK \cong \triangle JHK$	4.

What could be Reason 4?

- Ⓐ Side-Angle-Side congruence
- Ⓑ Hypotenuse-Leg congruence
- Ⓒ Angle-Angle-Side congruence
- Ⓓ Angle-Side-Angle congruence

Answer Key: D

Percentage of Students Answering Correctly: 35%

Reporting Category: Logic, Relationships, and Theorems

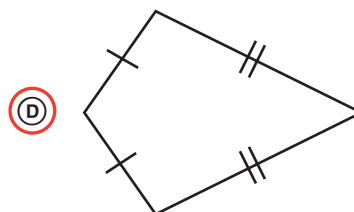
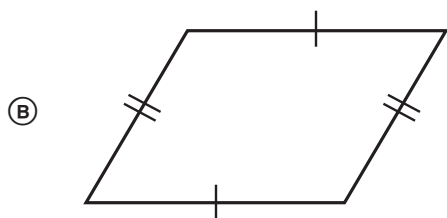
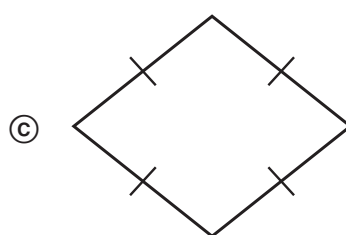
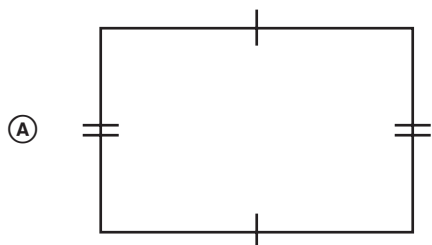
Benchmark: MA.912.GR.1.2

Benchmark Description: Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle and Hypotenuse-Leg.

42. Johanna makes the following claim:

"If a polygon is a quadrilateral, then two pairs of opposite sides are congruent."

Which polygon is a counterexample to Johanna's claim?



Answer Key: D

Percentage of Students Answering Correctly: 40%

Reporting Category: Logic, Relationships, and Theorems

Benchmark: MA.912.LT.4.10

Benchmark Description: Judge the validity of arguments and give counterexamples to disprove statements.

- 43.** A sequence of transformations maps triangle QRS onto congruent triangle XYZ .

Select all the sequences of transformations that could have mapped triangle QRS onto triangle XYZ .

- ☒ (A) a reflection across the x -axis, followed by a 90° clockwise rotation about the origin
- ☒ (B) a 180° clockwise rotation about the origin, followed by a reflection across the line $y = 3$
- ☐ (C) a dilation by a scale factor of 3 centered at the origin, followed by a translation 5 units down
- ☐ (D) a reflection across the line $y = x$, followed by a dilation by a scale factor of $\frac{1}{3}$ centered at the origin
- ☒ (E) a translation 1 unit left and 3 units up, followed by a 270° counterclockwise rotation about the origin

Answer Key: A, B, E

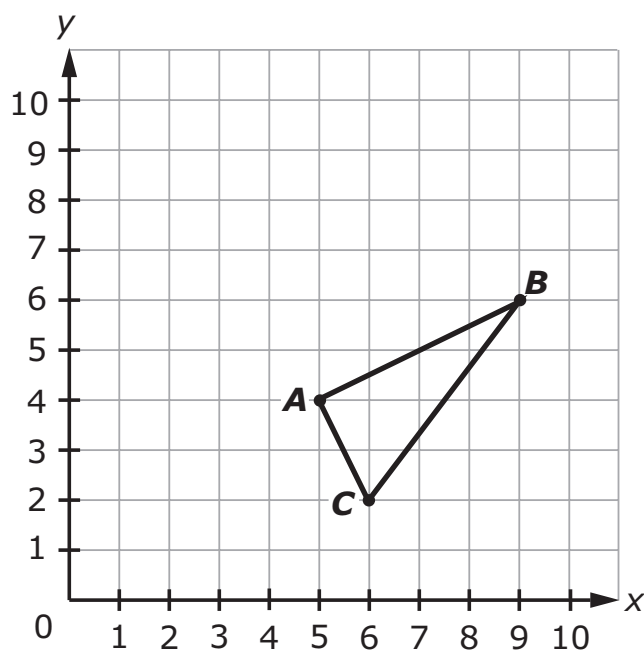
Percentage of Students Answering Correctly: 31%

Reporting Category: Congruence, Similarity, and Constructions

Benchmark: MA.912.GR.2.3

Benchmark Description: Identify a sequence of transformations that will map a given figure onto itself or onto another congruent or similar figure.

44. Triangle ABC is shown on the coordinate grid.



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Complete the steps to describe a sequence of transformations that will map triangle ABC onto itself.

Step 1	Translate triangle ABC 4 units up.															
Step 2	Translate triangle ABC 4 ← → ↶ ↷ ✕ <table border="1"><tr><td>1</td><td>2</td><td>3</td></tr><tr><td>4</td><td>5</td><td>6</td></tr><tr><td>7</td><td>8</td><td>9</td></tr><tr><td></td><td>0</td><td></td></tr><tr><td>.</td><td>-</td><td>$\frac{\Box}{\Box}$</td></tr></table> units down.	1	2	3	4	5	6	7	8	9		0		.	-	$\frac{\Box}{\Box}$
1	2	3														
4	5	6														
7	8	9														
	0															
.	-	$\frac{\Box}{\Box}$														
Step 3	Dilate triangle ABC by a scale factor of 1 ← → ↶ ↷ ✕ <table border="1"><tr><td>1</td><td>2</td><td>3</td></tr><tr><td>4</td><td>5</td><td>6</td></tr><tr><td>7</td><td>8</td><td>9</td></tr><tr><td></td><td>0</td><td></td></tr><tr><td>.</td><td>-</td><td>$\frac{\Box}{\Box}$</td></tr></table> centered at the origin.	1	2	3	4	5	6	7	8	9		0		.	-	$\frac{\Box}{\Box}$
1	2	3														
4	5	6														
7	8	9														
	0															
.	-	$\frac{\Box}{\Box}$														

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Answer Key: 4, 1, or any equivalent values

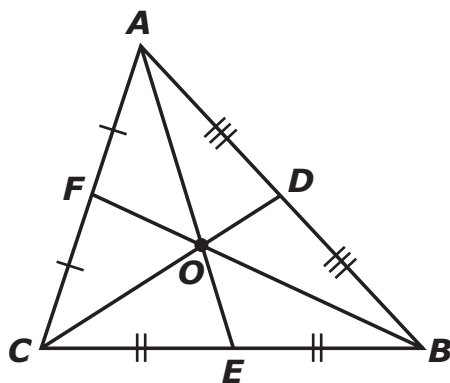
Percentage of Students Answering Correctly: 26%

Reporting Category: Congruence, Similarity, and Constructions

Benchmark: MA.912.GR.2.3

Benchmark Description: Identify a sequence of transformations that will map a given figure onto itself or onto another congruent or similar figure.

45. A figure is shown.



Complete the statement to describe $\triangle ABC$.

The [☒ A] medians [☐ B] angle bisectors [☐ C] perpendicular bisectors] of $\triangle ABC$ meet at point [☐ A] C. [☐ B] D. [☒ C] O.]

Answer Key: A, C

Percentage of Students Answering Correctly: 26%

Reporting Category: Logic, Relationships, and Theorems

Benchmark: MA.912.GR.1.3

Benchmark Description: Prove relationships and theorems about triangles. Solve mathematical and real-world problems involving postulates, relationships and theorems of triangles.